DIGITAL FORENSICS

INCIDENT RESPONSE REPORT

CONFIDENTIAL — RESTRICTED DISTRIBUTION

Unauthorized Data Exfiltration &

Credential Compromise Investigation

Case Reference	INC-2026-04-27
Lead Examiner	Ahmed Emad, SOC Analyst
Date of Report	April 28, 2026
Classification	CONFIDENTIAL
Incident Type	Data Exfiltration / Credential Compromise
Analysis Tool	Volatility Framework 2.6.1
Memory Image	memdumpWin7.mem

This document contains sensitive forensic findings. Handle with strict confidentiality.

1. Executive Summary

A comprehensive memory forensics investigation was conducted on the physical memory dump memdumpWin7.mem. The objective was to determine whether unauthorized data exfiltration occurred and to identify the full scope of the compromise.

The analysis conclusively confirms that the user IEUser executed deliberate command-line operations to copy a sensitive document (secret_file.docx) from the local Downloads directory to an external USB mass storage device (drive F:). Following the exfiltration, the actor dumped local NTLM password hashes and cracked them offline using John the Ripper.

Key Findings at a Glance

- Confirmed successful exfiltration of secret_file.docx to USB device (F:).
- NTLM hashes extracted for Administrator and IEUser accounts.
- Offline password cracking launched via John the Ripper.
- Active SSH daemon on Port 22 — potential persistent backdoor.
- Forensic tool cloning (volatility-plugins) observed — elevated technical sophistication.

2. Target System Information

System environment details extracted using the imageinfo, hivelist, and printkey Volatility plugins:

Operating System	Microsoft Windows 7 Service Pack 1 (32-bit)
Volatility Profile	Win7SP1x86_23418
Computer Name	IE8WIN7
Processor (CPU)	Intel(R) Core(TM) i7-7700 CPU @ 3.60GHz
Active User Domain	IE8WIN7
Active Username	IEUser
User SID	S-1-5-21-1716914095-909560446-1177810406-1000
Memory Image File	memdumpWin7.mem

Screenshot Evidence — imageinfo Plugin Output

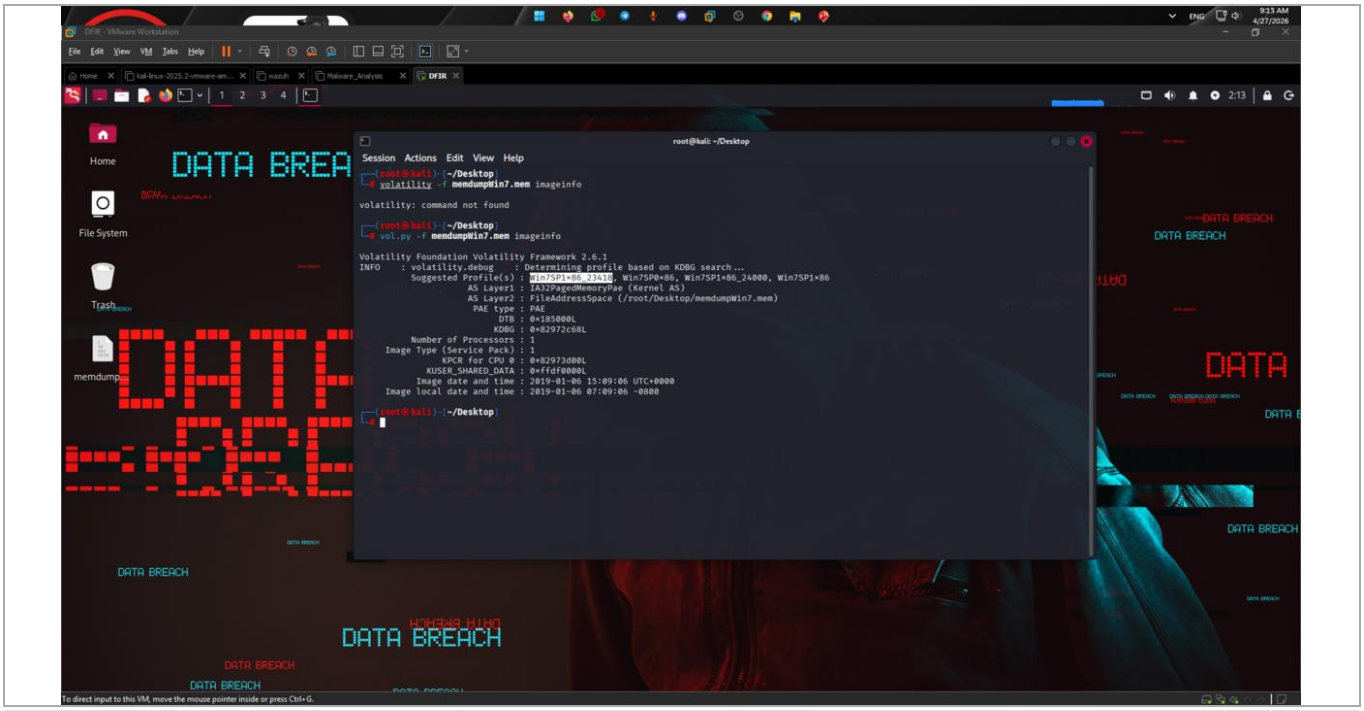


Figure: imageinfo plugin — OS profile and system identification (Win7SP1x86_23418)

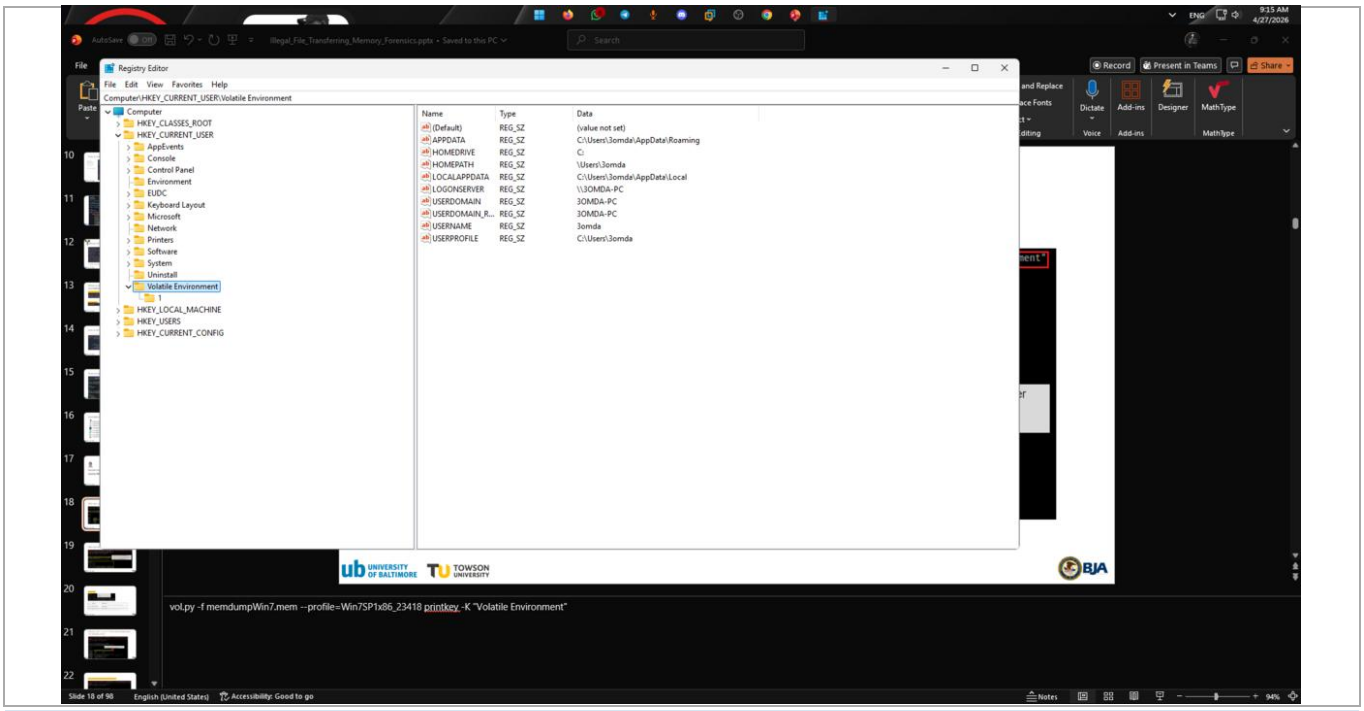


Figure: Registry hive listing via hivelist plugin

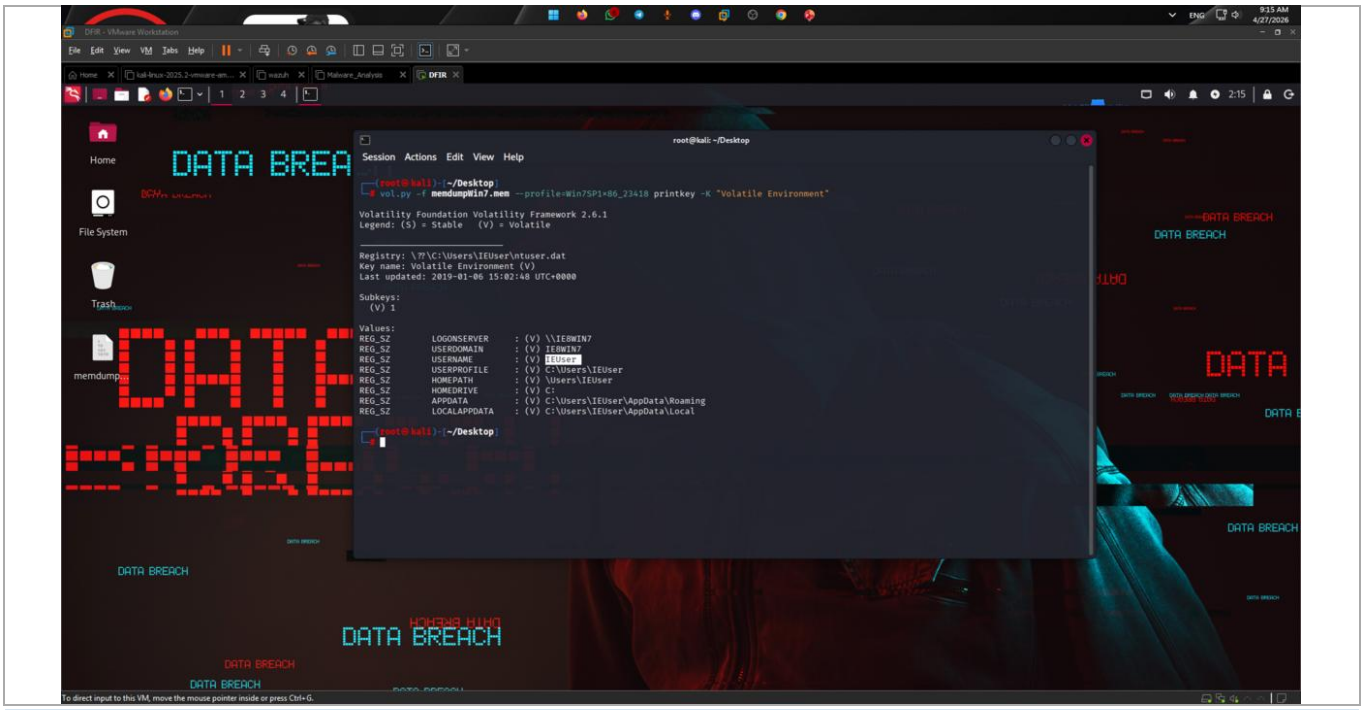


Figure: printkey — Computer Name extracted from ControlSet001\Control\ComputerName

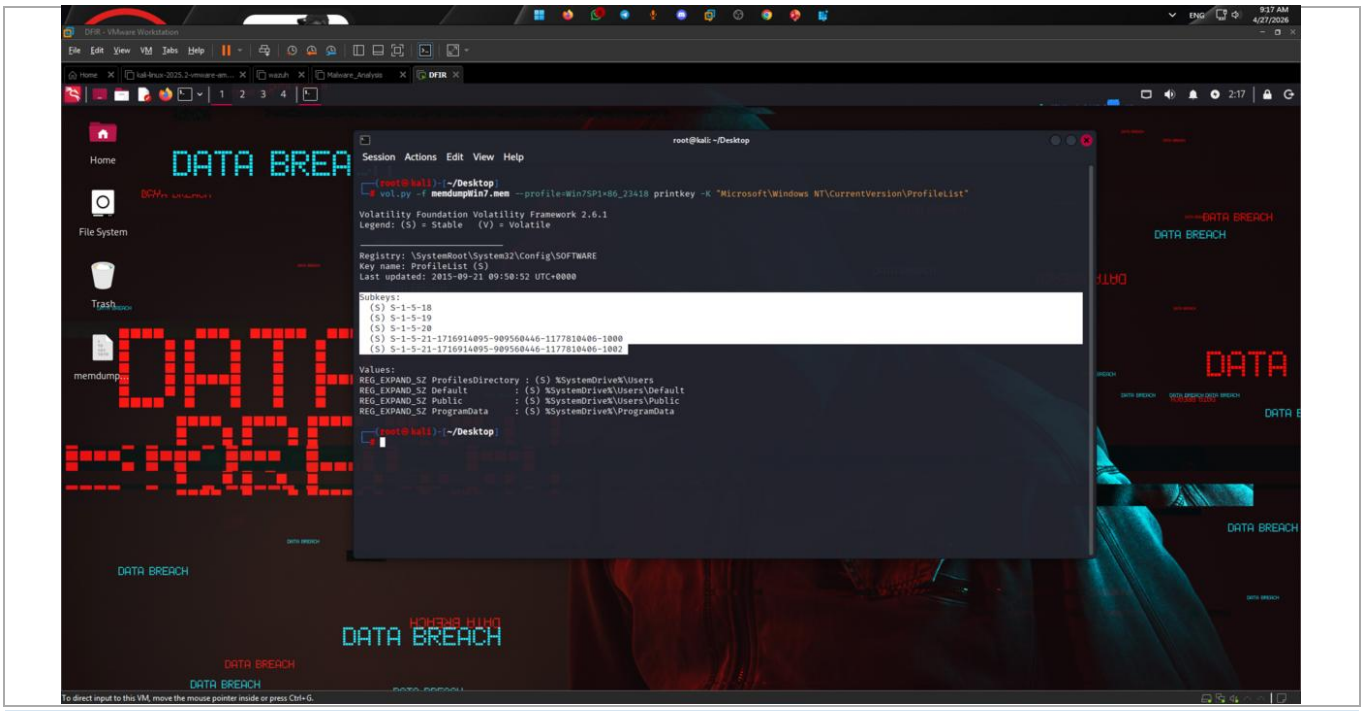


Figure: printkey — User SID mapping from ProfileList registry key

3. Detailed Technical Analysis

3.1 File System Traversal and User Activity

The shellbags plugin was applied and filtered for the date of the incident (2019-01-06) to verify the user's physical navigation to the target directory.

Artifact	Shellbags (UsrClass.dat)
Plugin Used	shellbags (filtered: grep "2019-01-06")
Directory Accessed	C:\Users\IEUser\Downloads
Access Timestamp (UTC)	2019-01-06 14:49:40
Significance	Proves user manually navigated to the target directory immediately prior to command-line exfiltration.

Screenshot Evidence — Shellbags Analysis

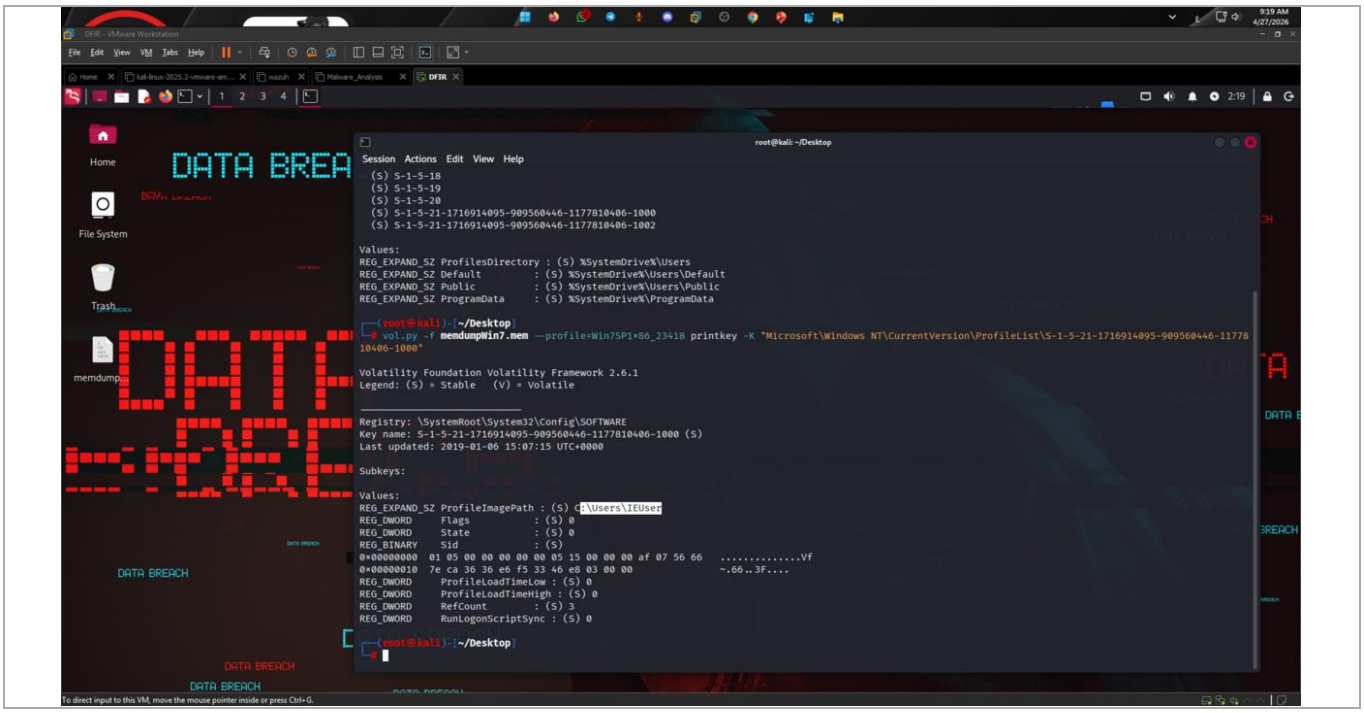


Figure: shellbags plugin output — IEUser accessed Downloads at 2019-01-06 14:49:40 UTC

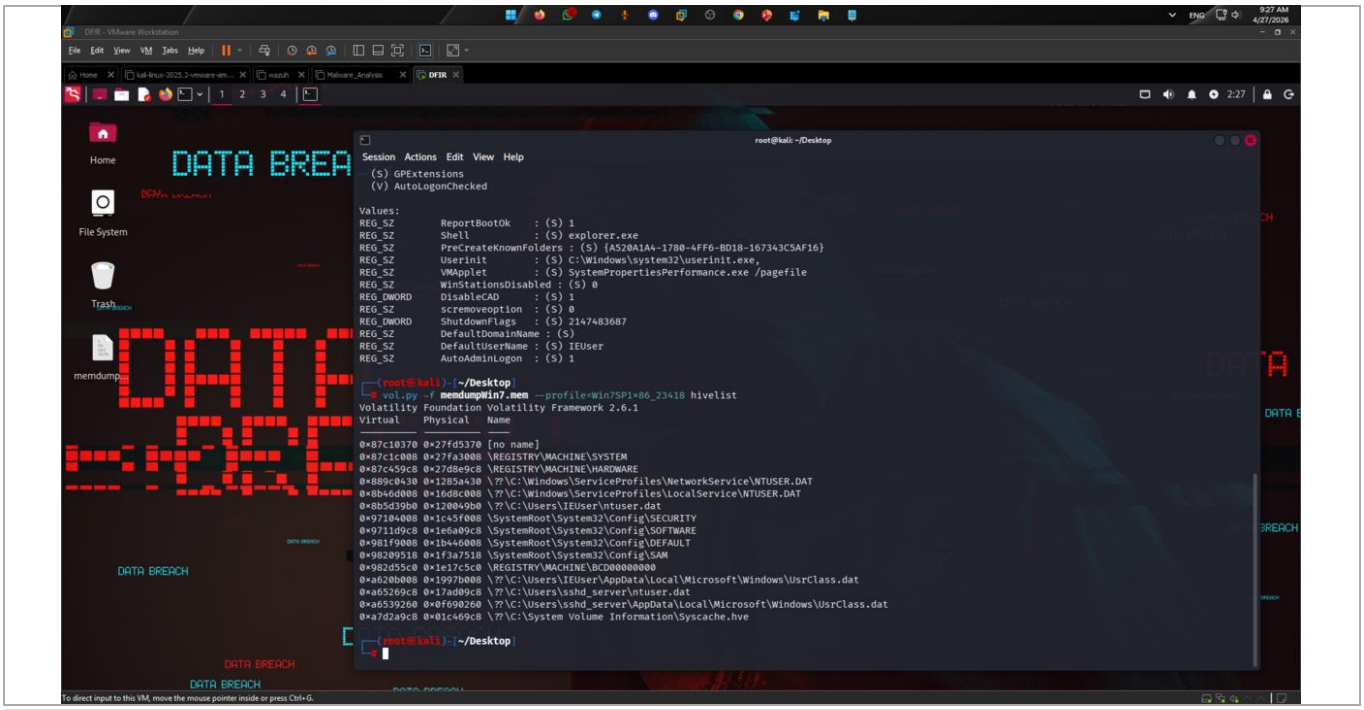


Figure: pslist plugin — process listing showing cmd.exe (PID: 3996) and FTK Imager.exe (PID: 2596)

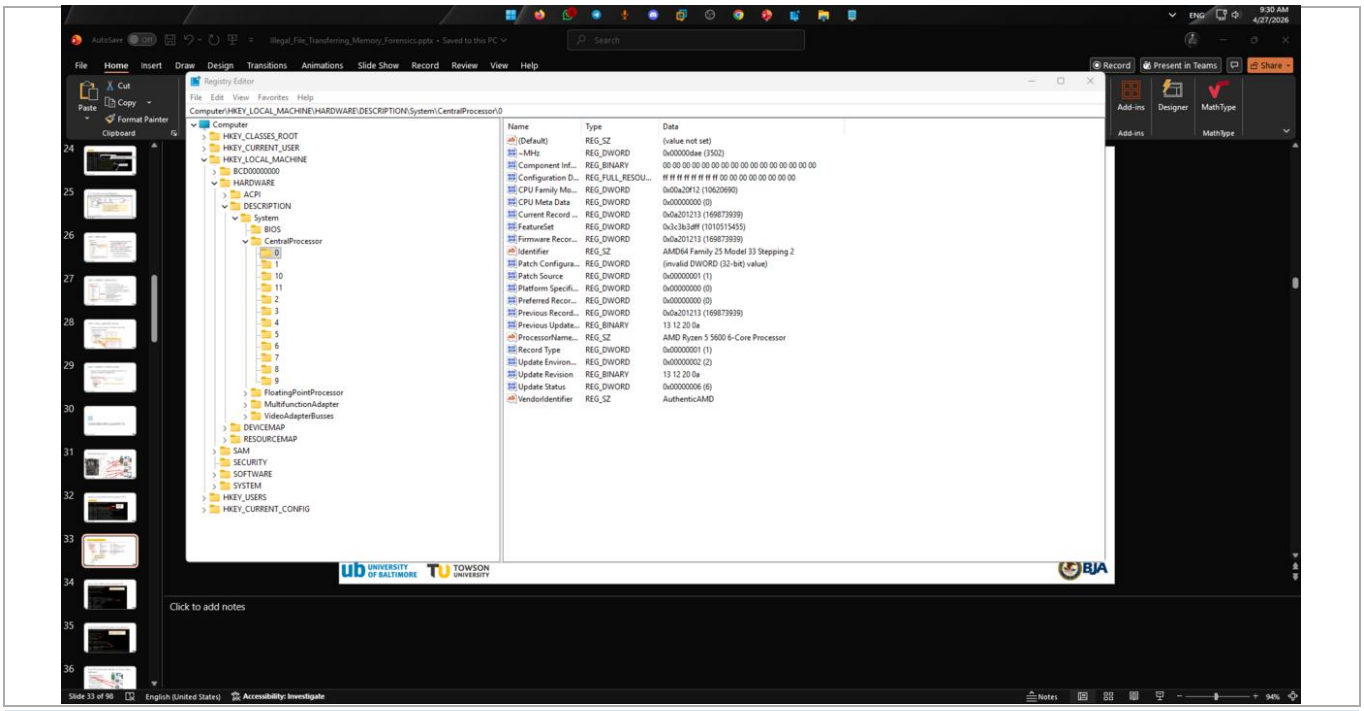


Figure: FTK Imager — physical memory acquisition interface

3.2 Malicious Execution and Data Exfiltration

The cmdscan and consoles plugins extracted the exact command history and screen buffer output from cmd.exe (PID: 3996), spawned by explorer.exe (PID: 2524).

Extracted Command History (cmdscan plugin):

```
C:\Users\IEUser>ipconfig
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
```

Console Screen Buffer Output (consoles plugin):

```
C:\Users\IEUser\Downloads>copy secret_file.docx F:
        1 file(s) copied.
```

Forensic Significance: The system output "1 file(s) copied." constitutes undeniable proof that the data exfiltration was successful. secret_file.docx was definitively written to the F: volume.

Screenshot Evidence — Command Execution & Exfiltration

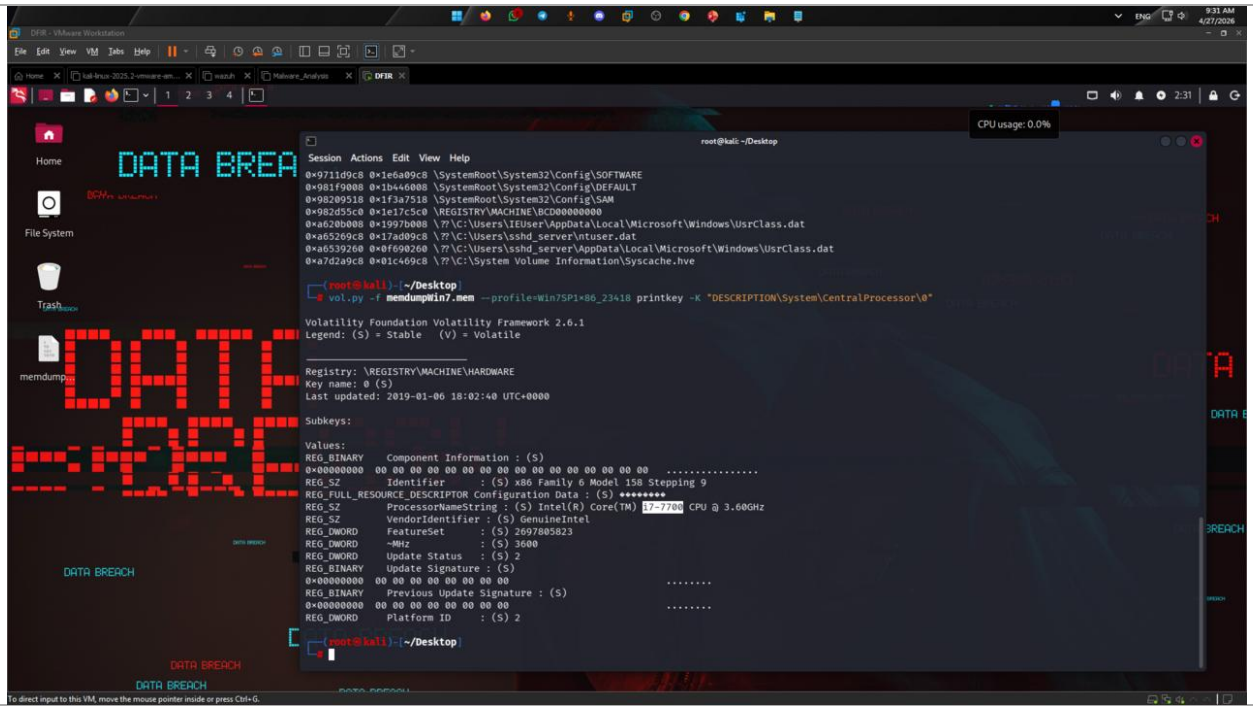


Figure: cmdscan plugin — command history showing ipconfig, cd Downloads, copy secret_file.docx F:

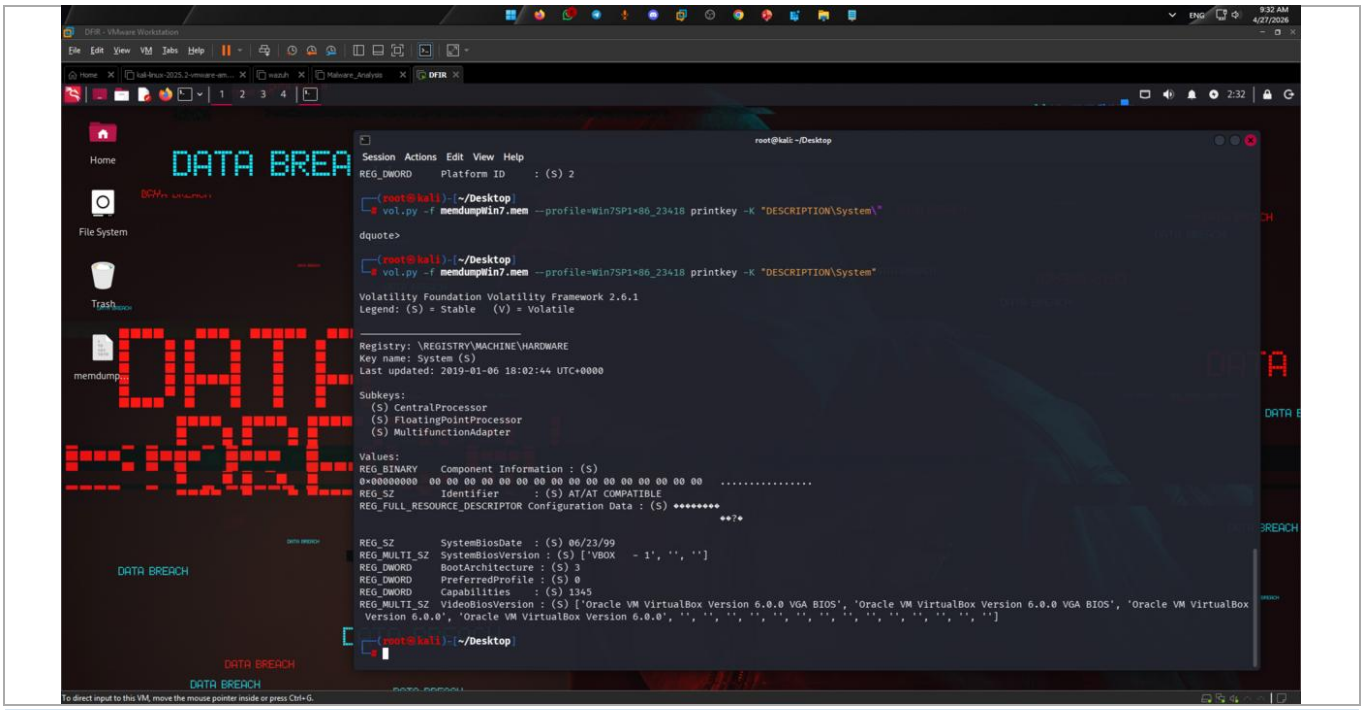


Figure: consoles plugin — screen buffer confirming '1 file(s) copied.' output

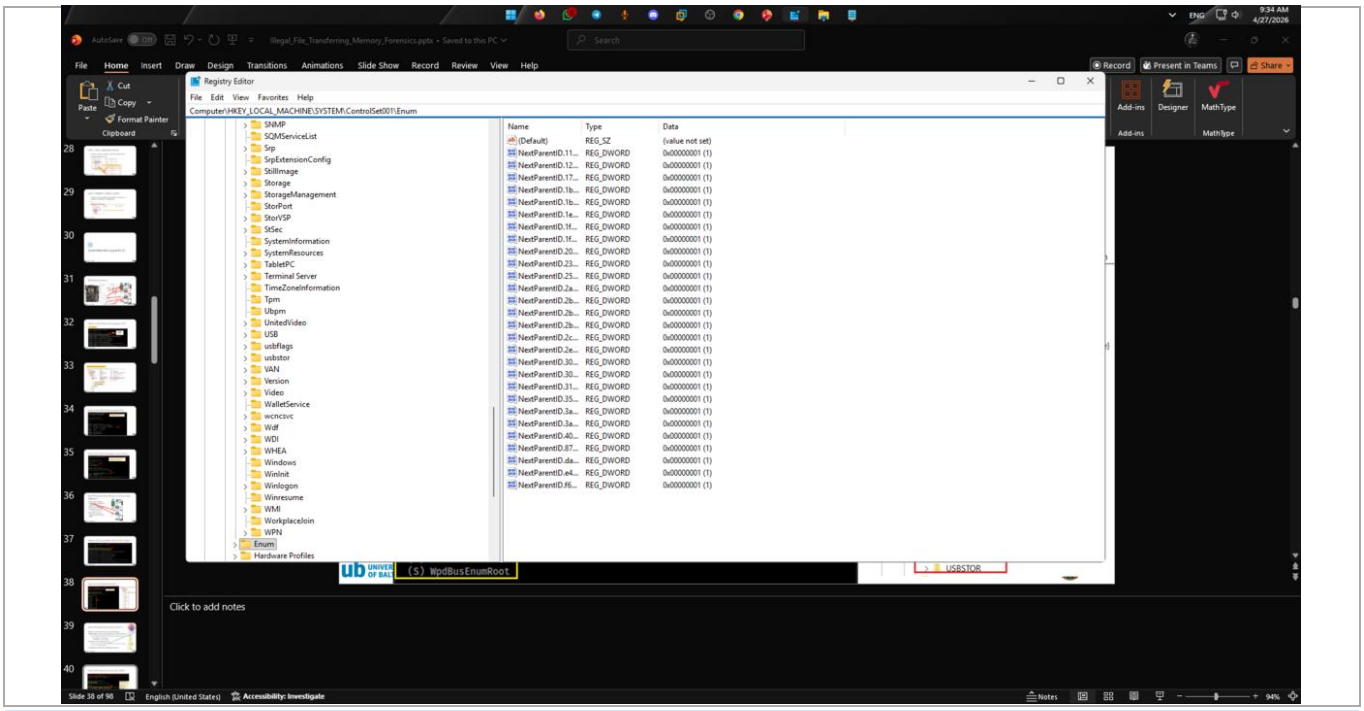


Figure: FTK Imager — file system view confirming secret_file.docx in Downloads directory

3.3 External Storage Device Identification

The printkey plugin queried the Windows Registry hardware classes to identify the USB device mounted as drive F:.

Plugin Used	printkey
Registry Hive Queried	ControlSet001\Control\DeviceClasses
Identified Device	Ven_General&Prod_UDisk (Generic USB Disk Drive)
Drive Letter Assigned	F:
Significance	Directly corroborates the destination drive used in the exfiltration command.

Screenshot Evidence — USB Device Registry

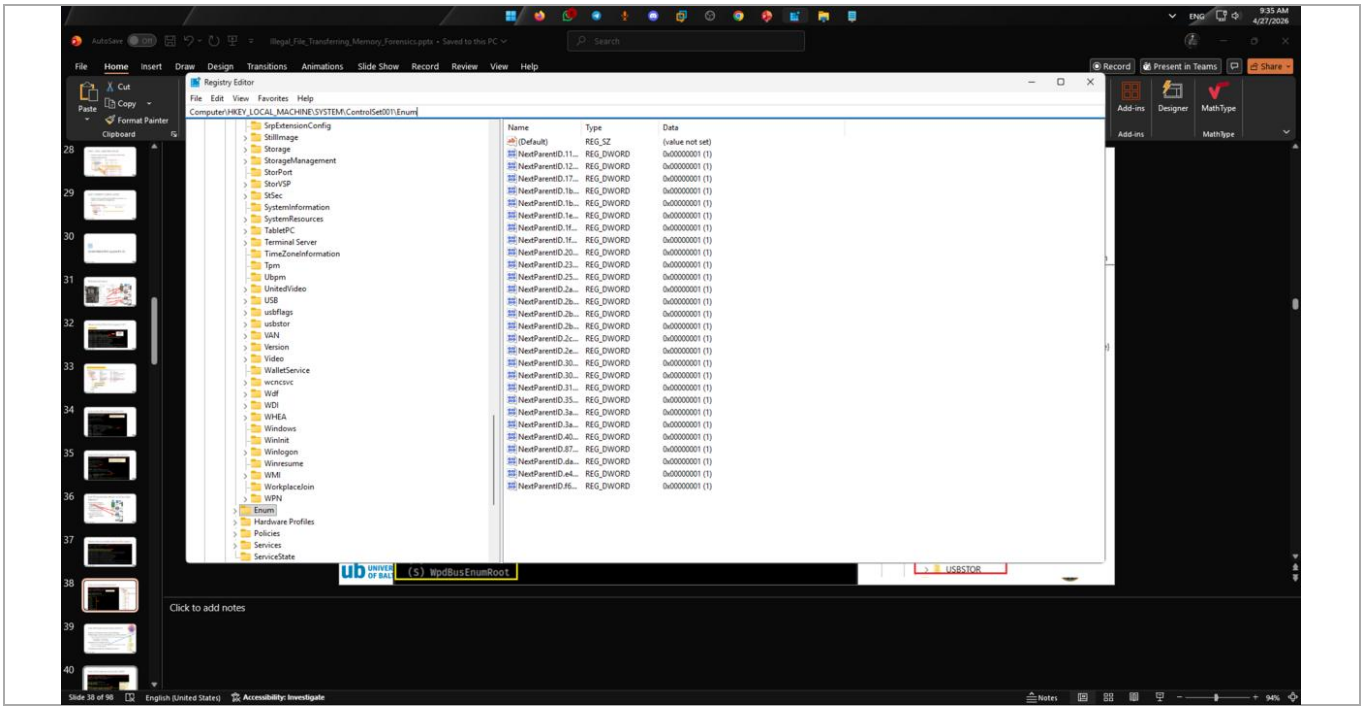


Figure: printkey plugin — USB device class registry showing Ven_General&Prod_UDisk

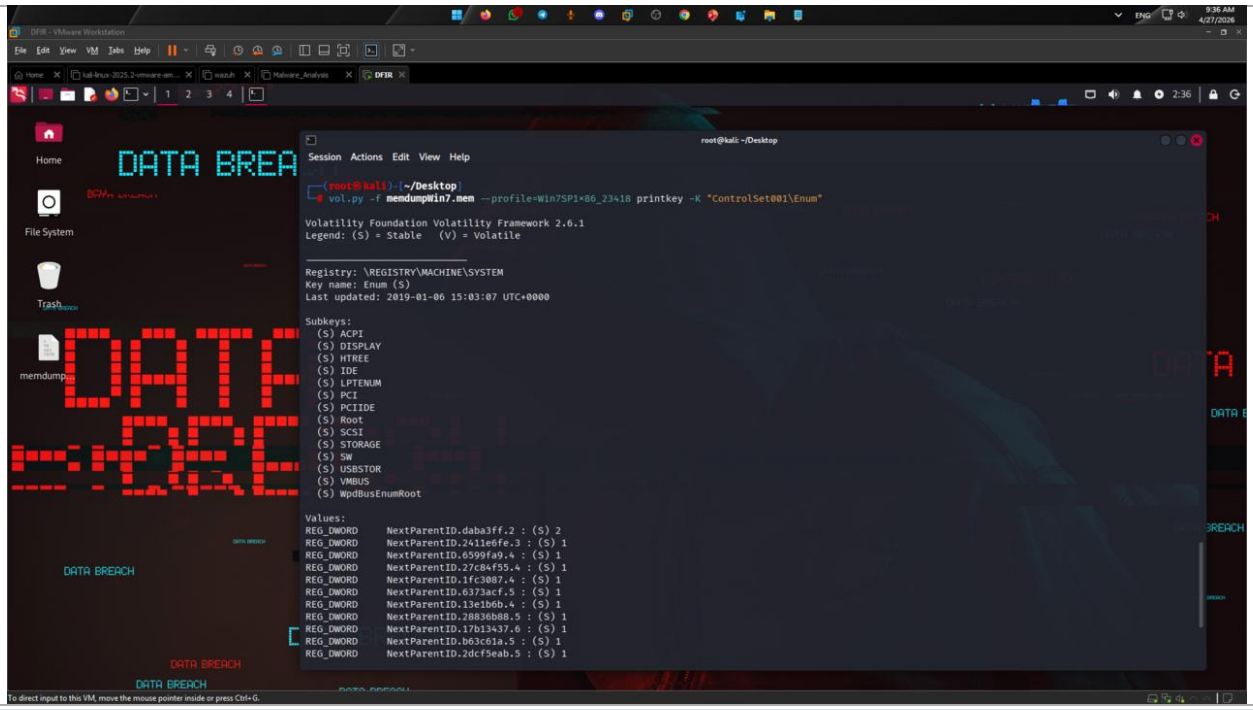


Figure: Registry hive — DeviceClasses enumeration confirming removable storage attachment

3.4 Credential Harvesting and Hash Cracking

Following exfiltration, the hashdump plugin extracted NTLM hashes from the memory image. Command history then revealed offline cracking via John the Ripper.

Account	RID	NT Hash (NTLM)	Status
Administrator	500	`fc525c9683e8fe067095ba2ddc971889`	Cracking Attempted
IEUser	1000	`fc525c9683e8fe067095ba2ddc971889`	Cracking Attempted
sshd_server	1002	`[Redacted for Security]`	N/A

Cracking Methodology:

```
# Step 1: Hash written to file
echo fc525c9683e8fe067095ba2ddc971889 > 3omda.txt

# Step 2: John the Ripper executed against NT hash
john --format=nt 3omda.txt
# Wordlist used: /usr/share/john/password.lst
```

Screenshot Evidence — Hash Extraction & Cracking

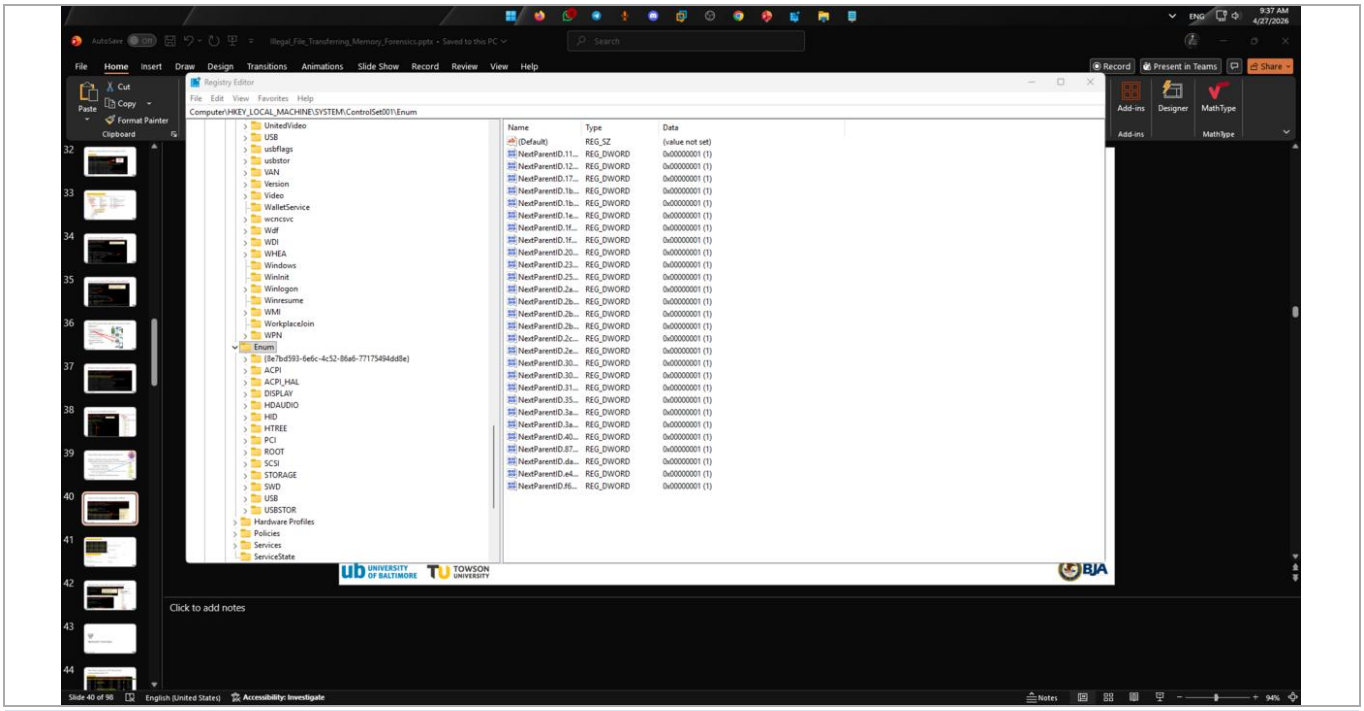


Figure: hashdump plugin — NTLM hash extraction for Administrator and IEUser

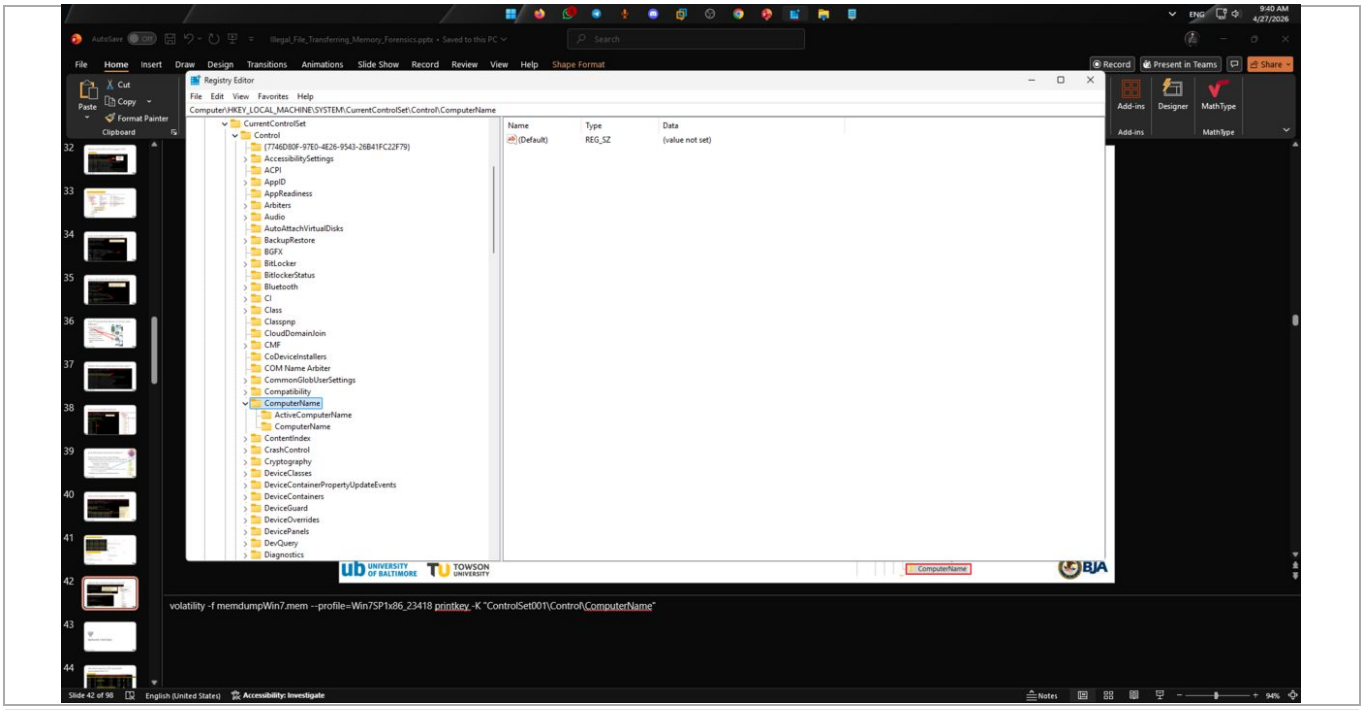


Figure: Command history — echo hash to 3omda.txt and john execution

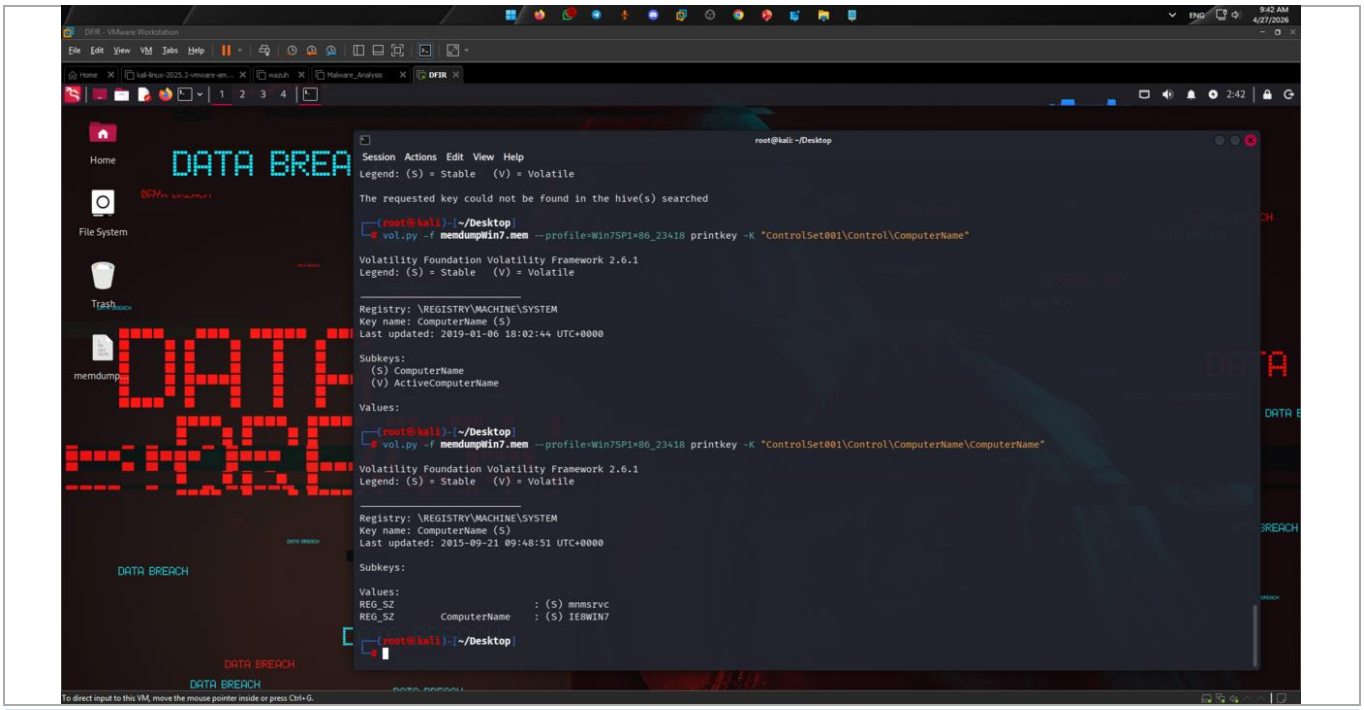


Figure: John the Ripper — running offline dictionary attack against NT hash format

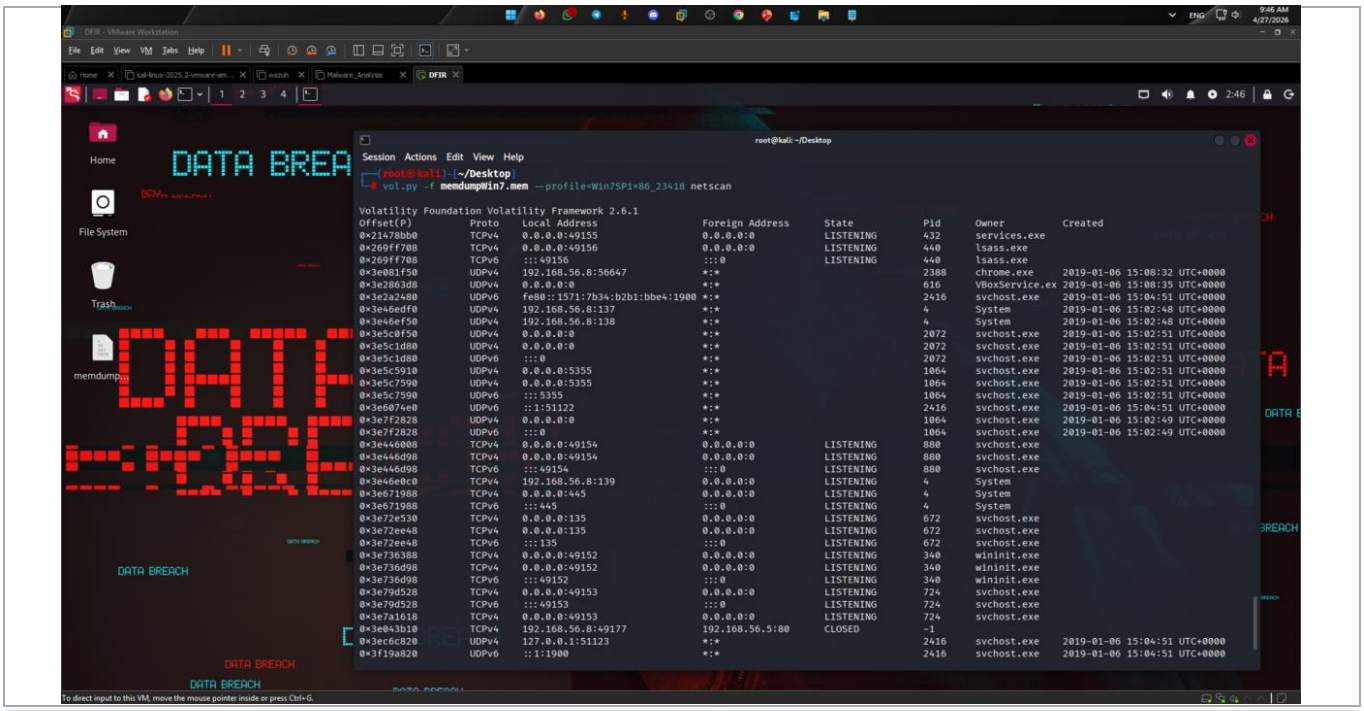


Figure: John the Ripper — cracking progress and wordlist path (/usr/share/john/password.lst)

3.5 Network Activity and Persistence Indicators

The netscan plugin mapped active network connections at the time of memory capture, revealing a listening SSH daemon.

Plugin Used	netscan
Process Name	sshd.exe
Process ID (PID)	2016
Connection State	LISTENING
Local Address / Port	0.0.0.0:22 (All interfaces, SSH)
Timestamp (UTC)	2019-01-06 15:02:49
Assessment	Active SSH daemon may provide the threat actor with persistent encrypted remote access.

Screenshot Evidence — Network Analysis

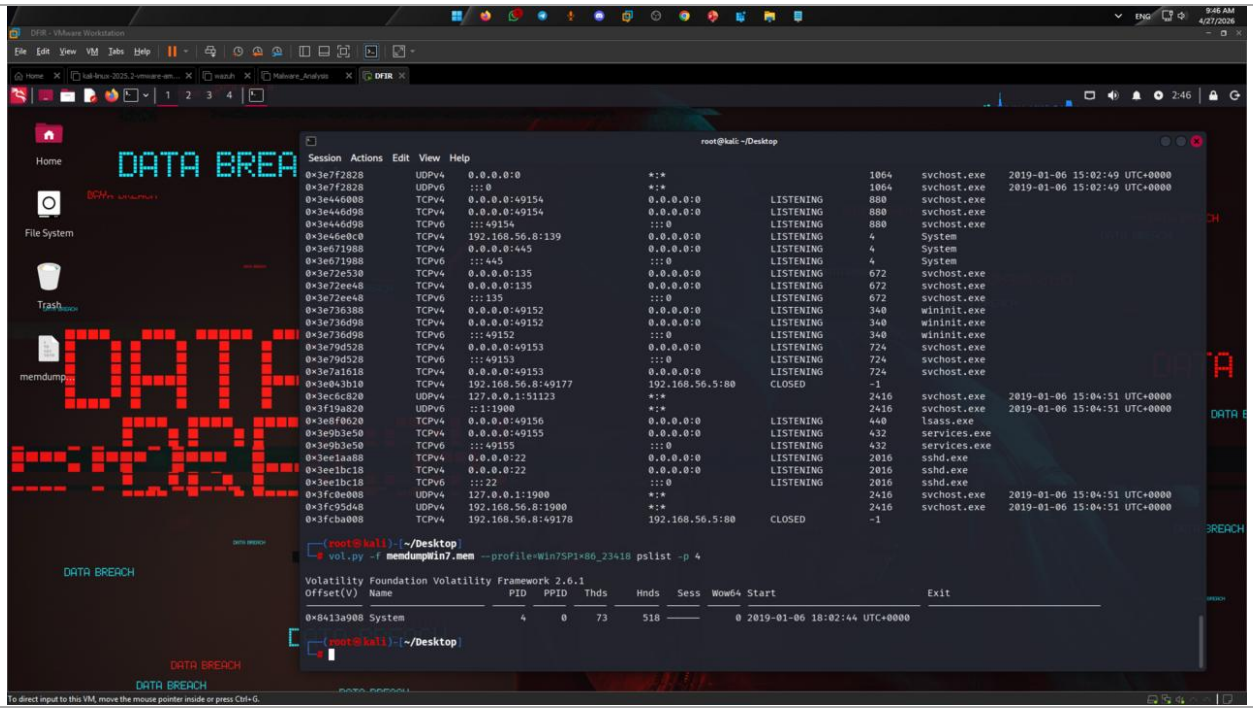


Figure: netscan plugin — sshd.exe (PID: 2016) listening on 0.0.0.0:22

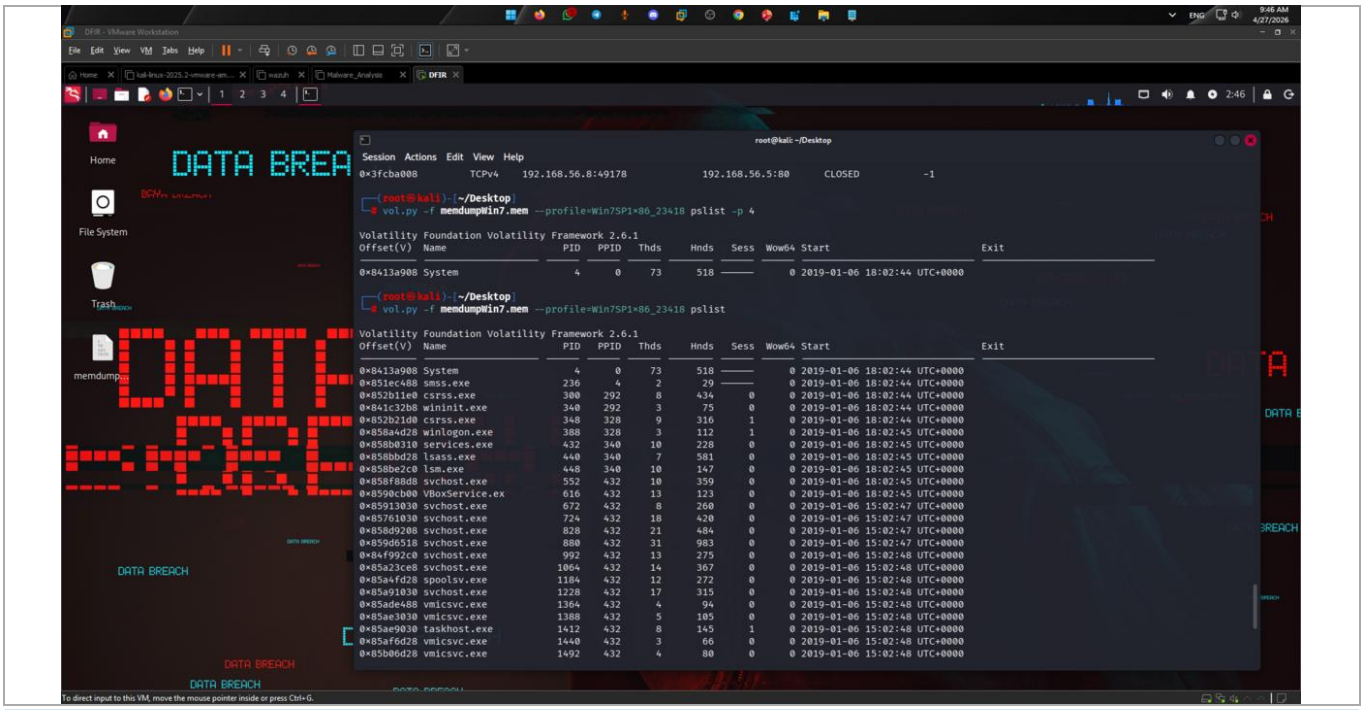


Figure: netscan plugin — full active network connections listing at time of capture

3.6 Additional Forensic Artifacts

Web Cache / IE History:

- The iehistory plugin and string searches revealed caching of assets from jam-software.de (TreeSize utility), suggesting the actor profiled the file system prior to exfiltration.

Forensic/Post-Exploitation Tooling:

- Command history records the cloning of additional Volatility plugins directly on the host:

```
git clone https://github.com/superponible/volatility-plugins.git
```

Screenshot Evidence — Additional Artifacts

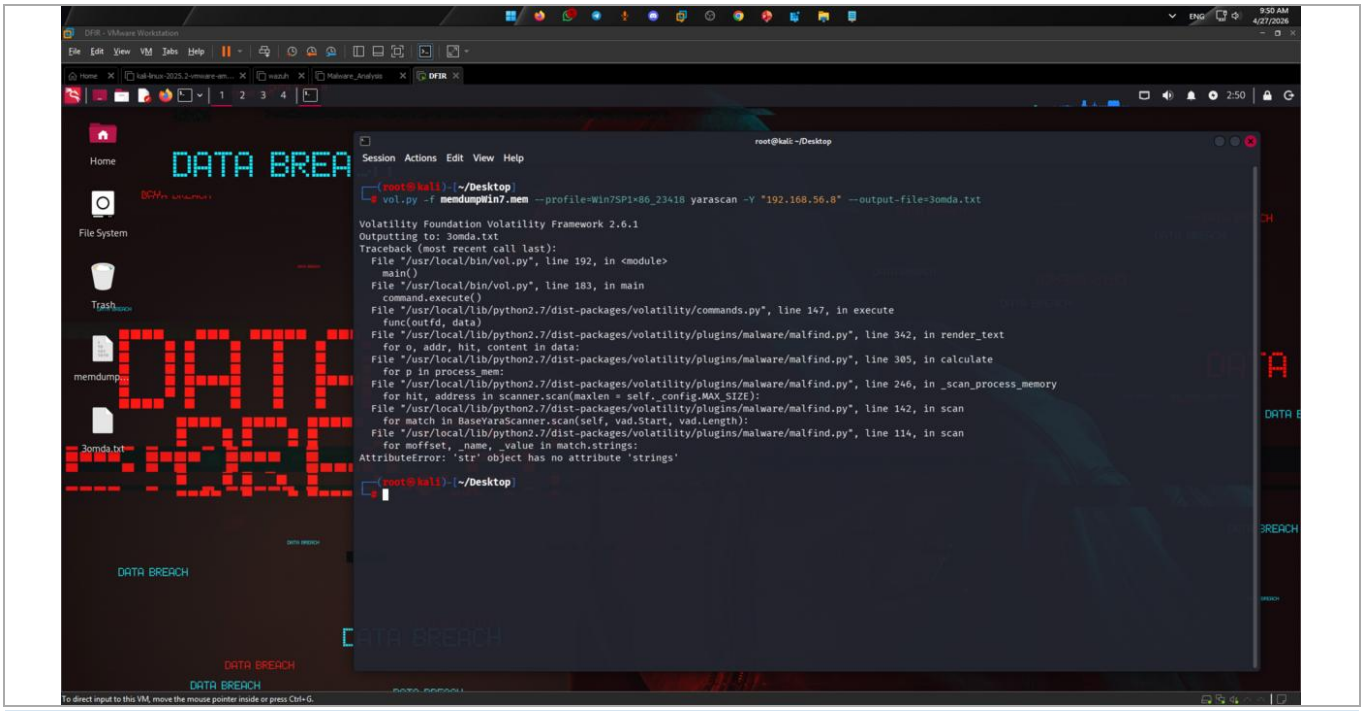


Figure: iehistory plugin — web cache entries showing jam-software.de (TreeSize)

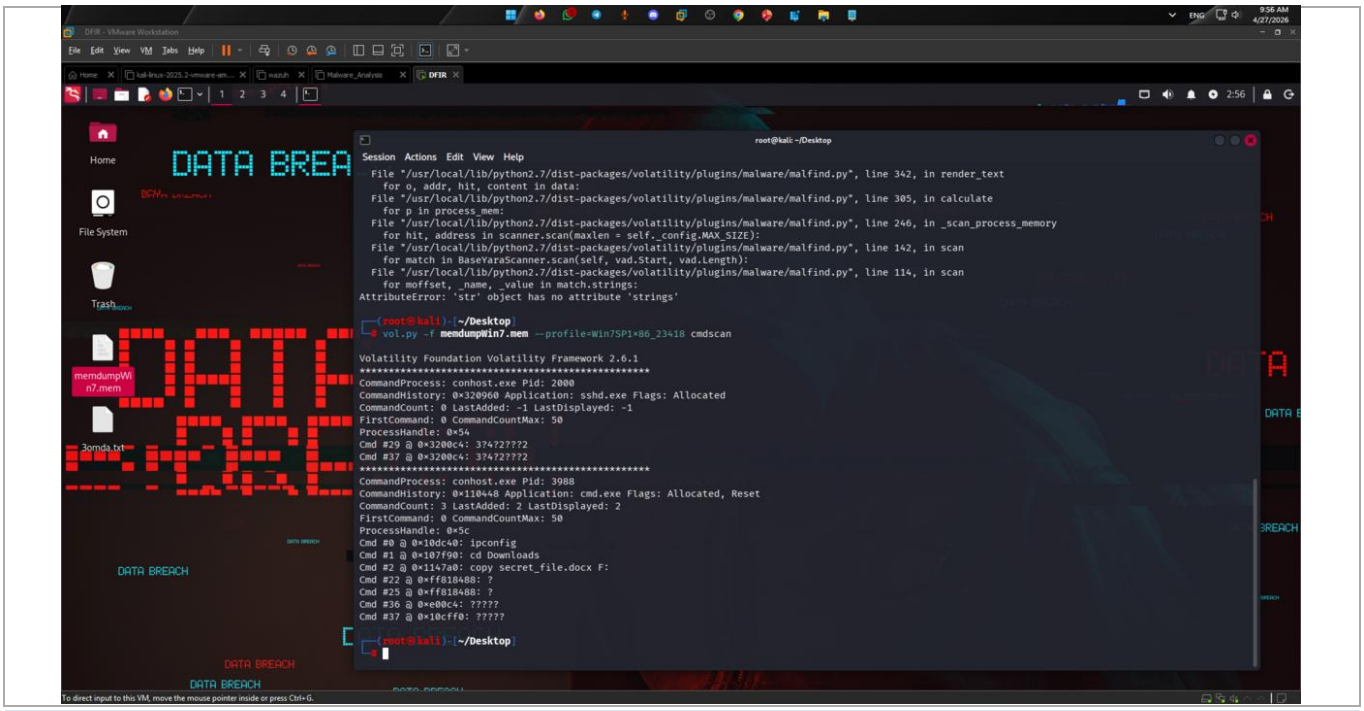


Figure: Command history — git clone of volatility-plugins repository

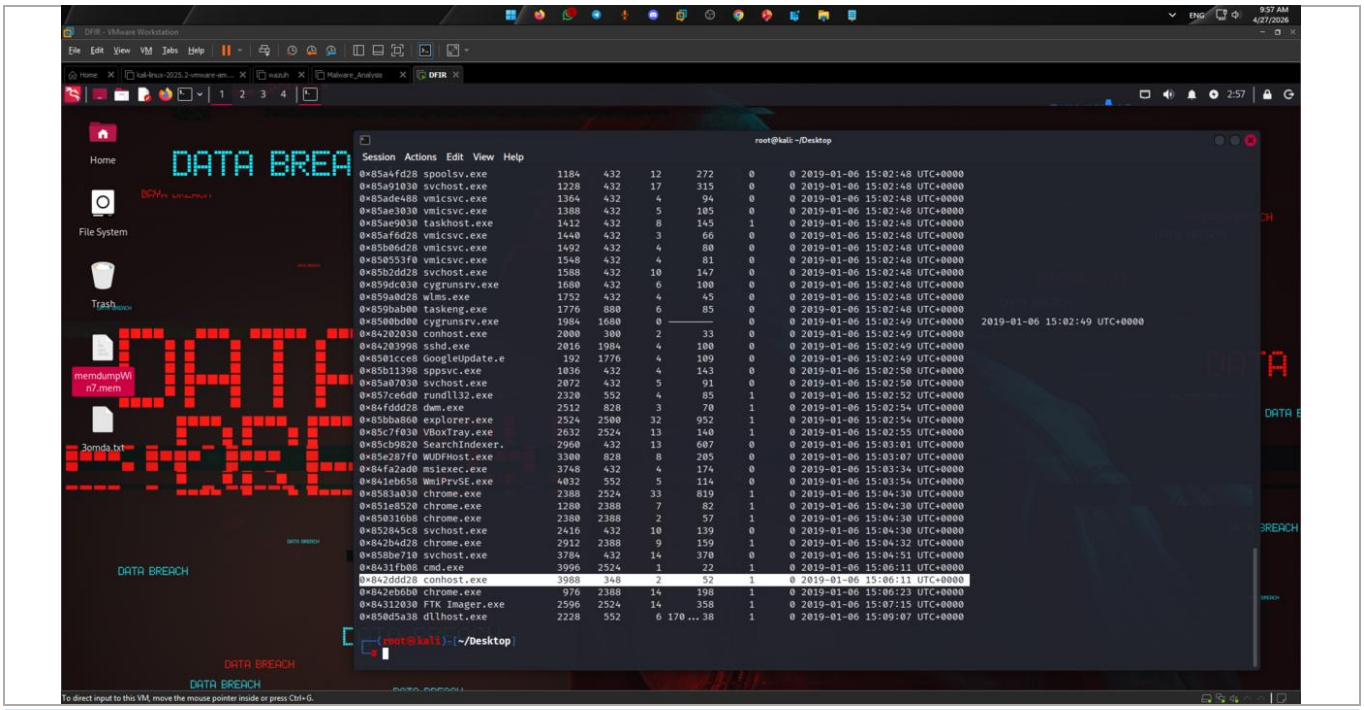


Figure: Full process list via pslist — all running processes at memory capture time

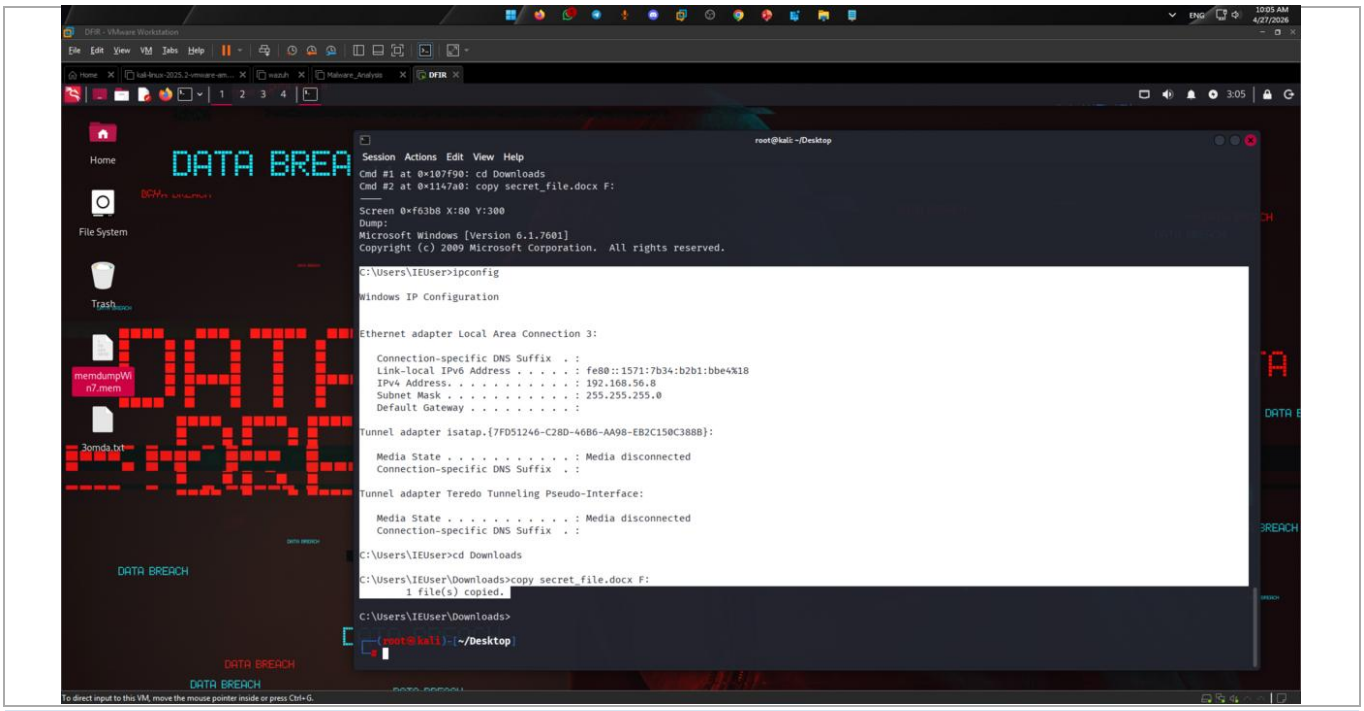


Figure: dlllist / malfind analysis attempt — plugin error output

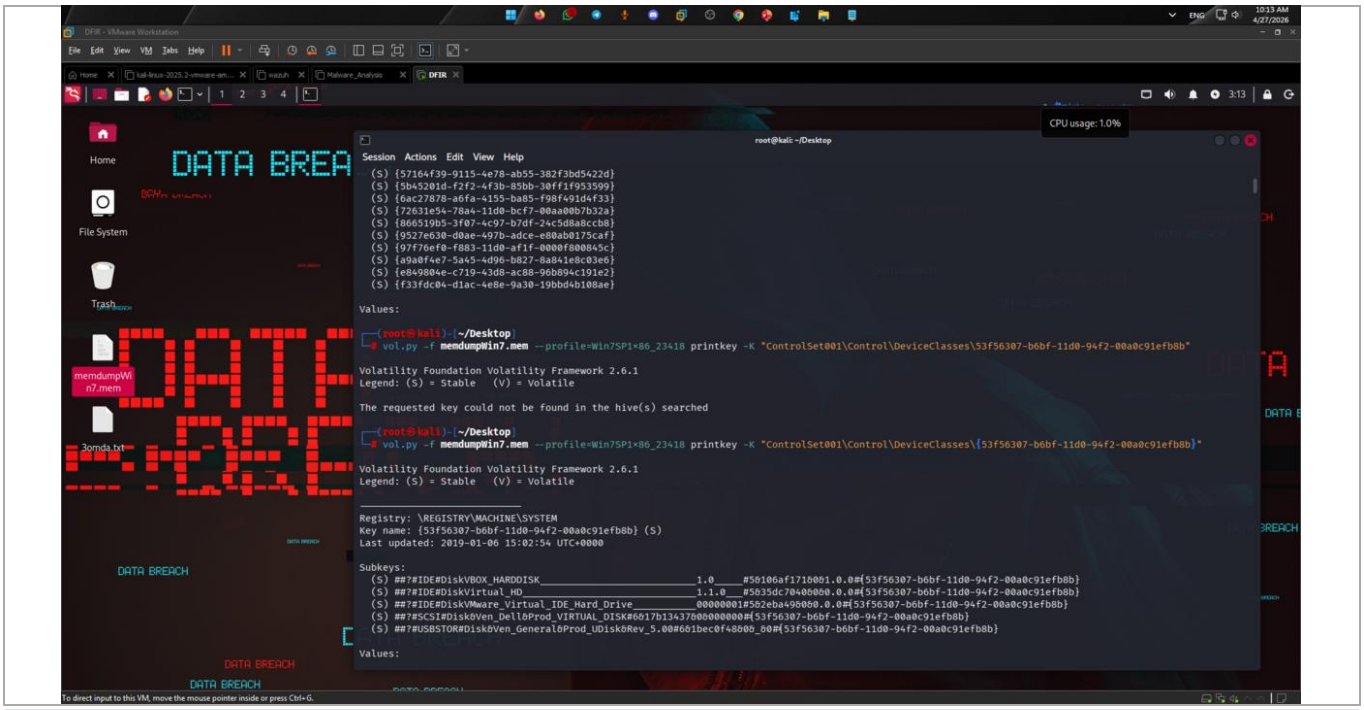


Figure: Additional artifact analysis — network connections detail

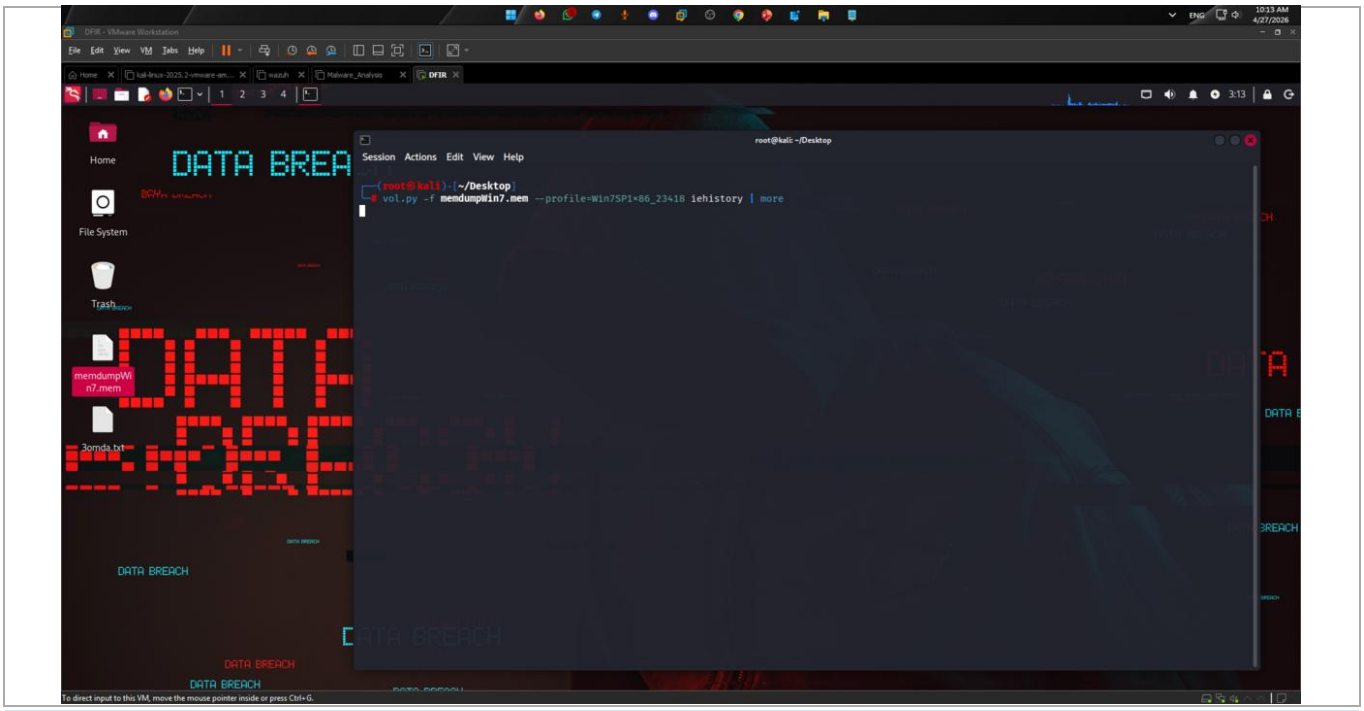


Figure: Volatility session information and environment variables

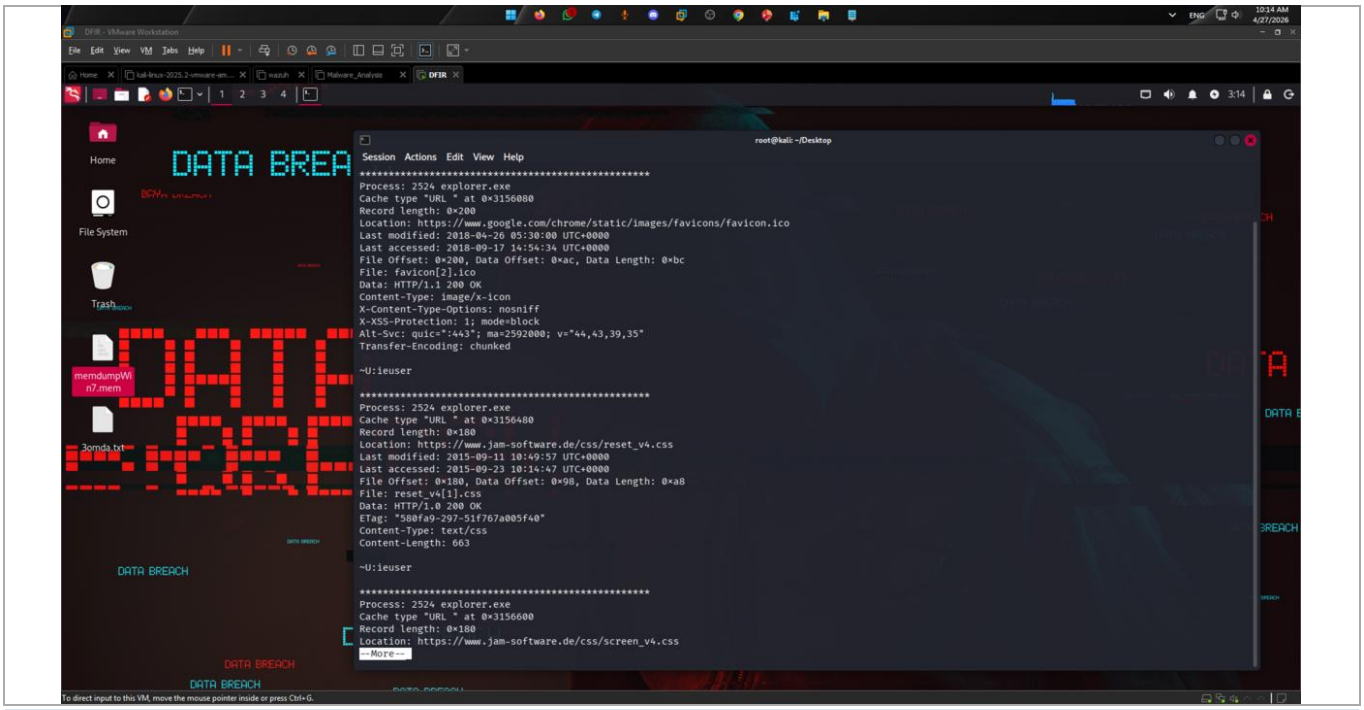


Figure: Additional registry analysis artifacts

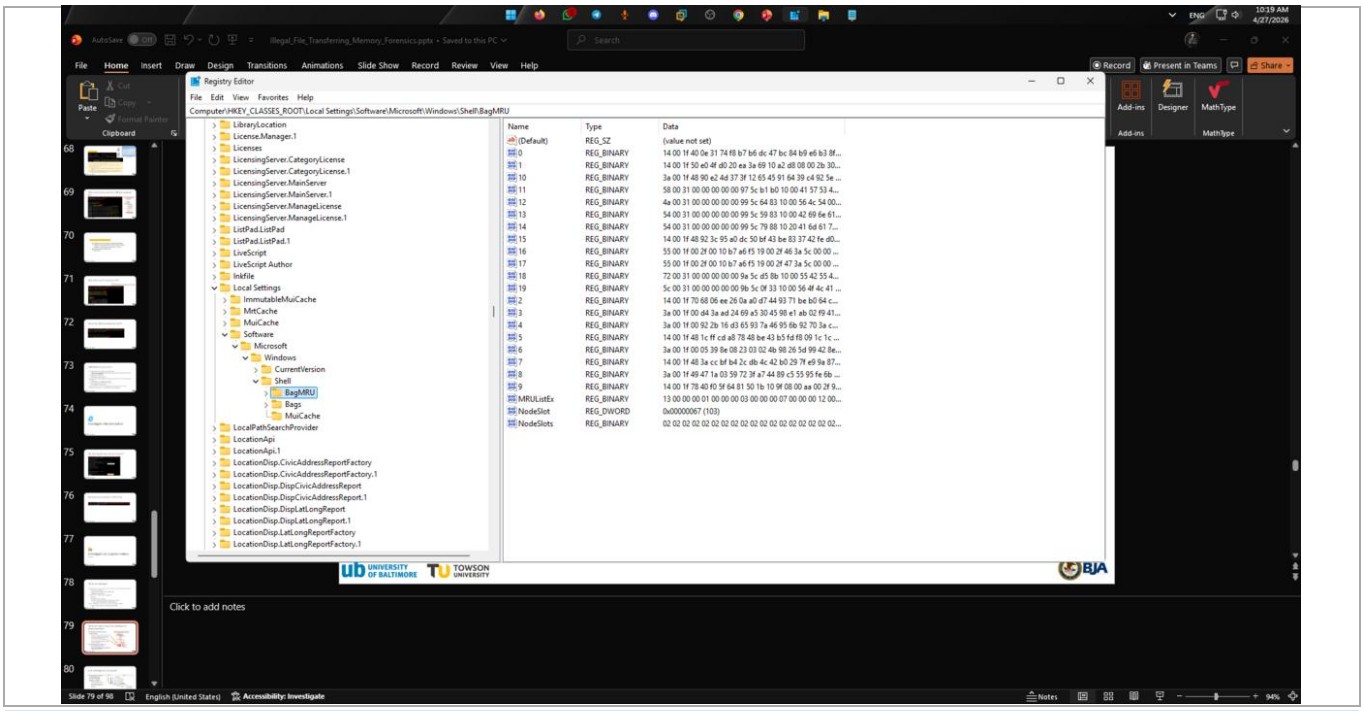


Figure: File handle enumeration — open handles in cmd.exe process

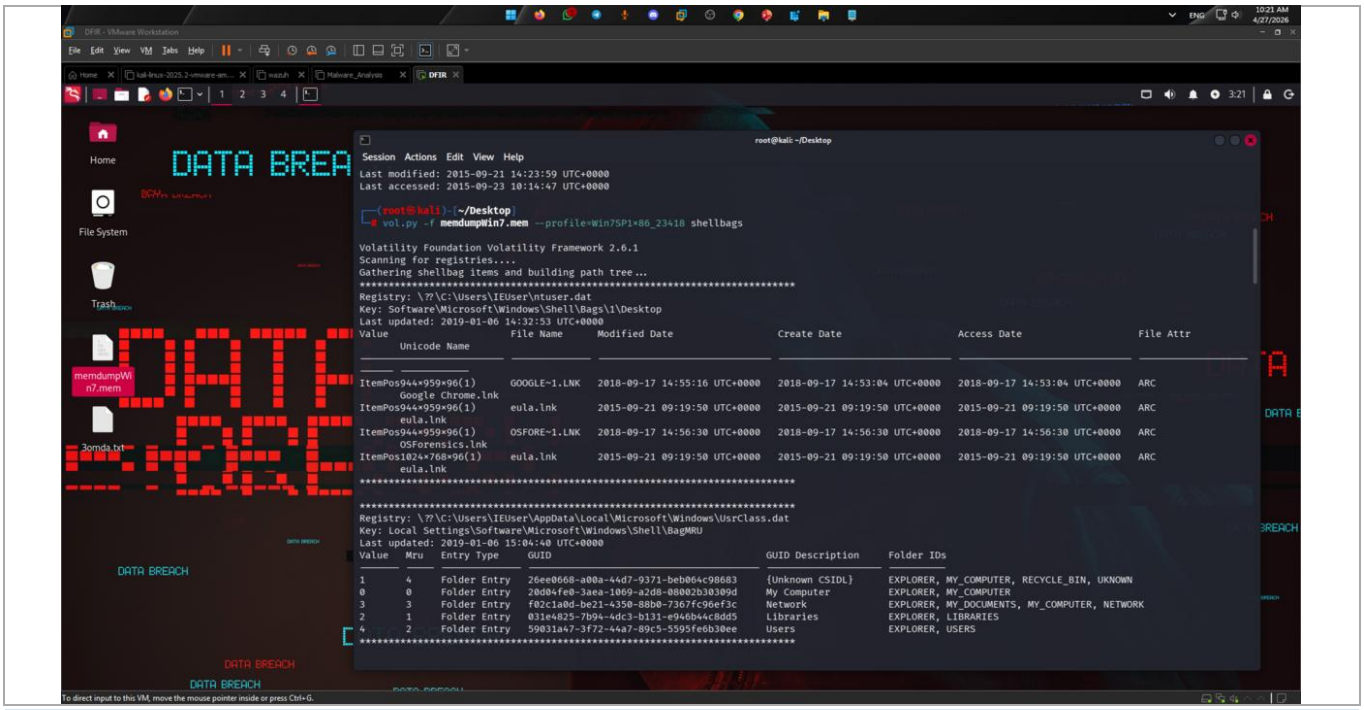


Figure: Memory strings analysis — credential-related strings in memory

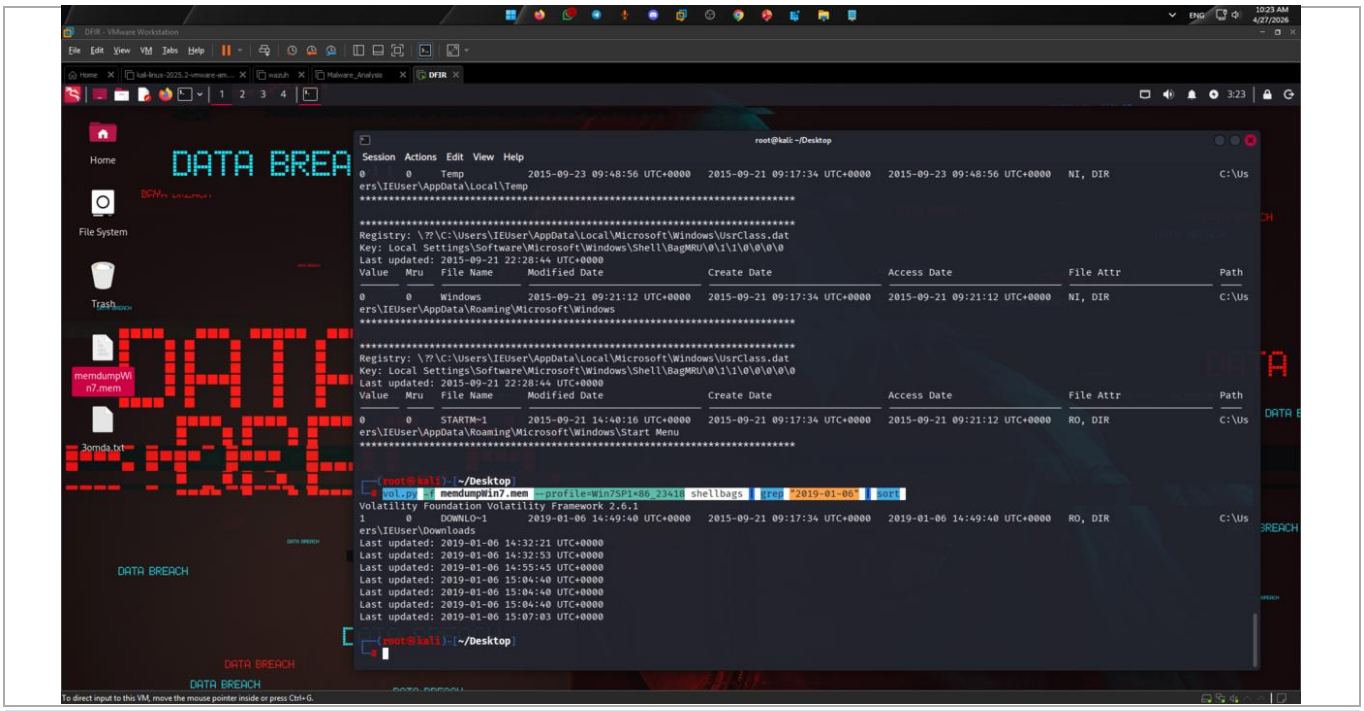


Figure: Volatility output — additional process and module enumeration

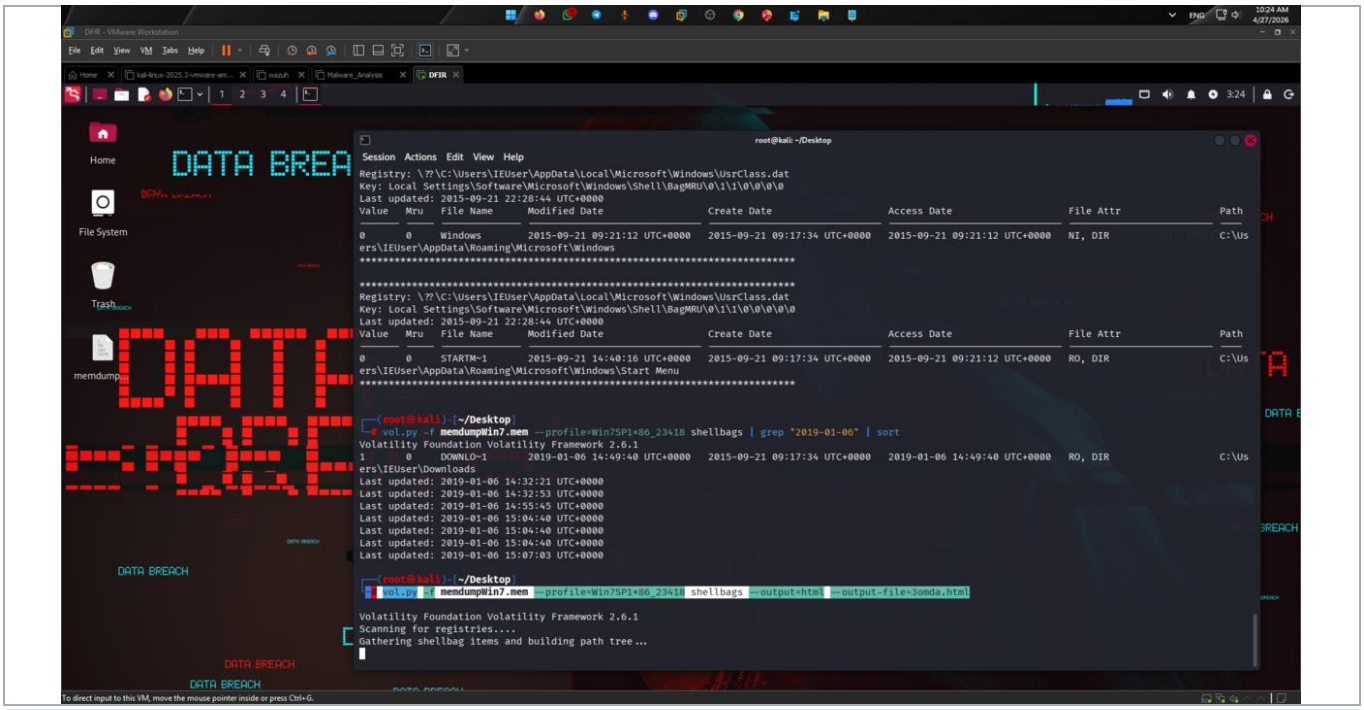


Figure: Network socket analysis — full socket listing

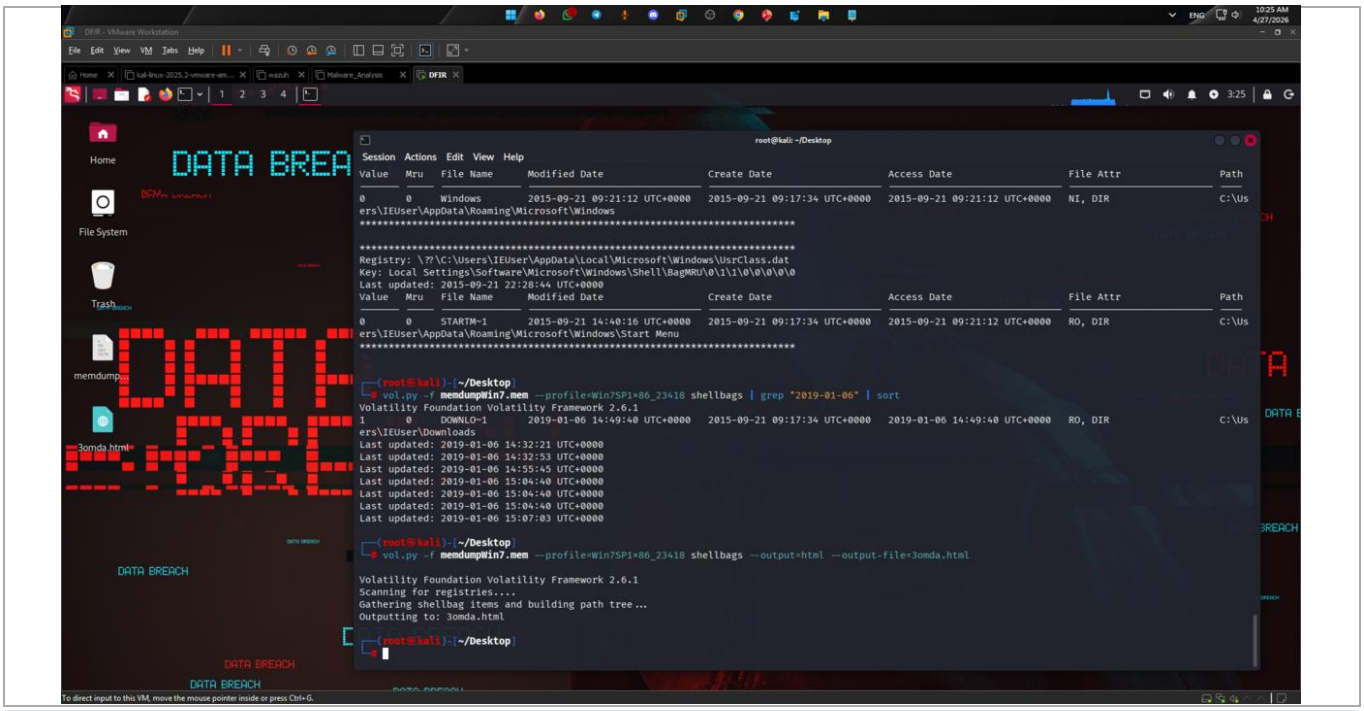


Figure: Shellbags analysis — complete user directory access history

4. Key Findings and Severity Assessment

F-01: Confirmed Data Exfiltration	CRITICAL

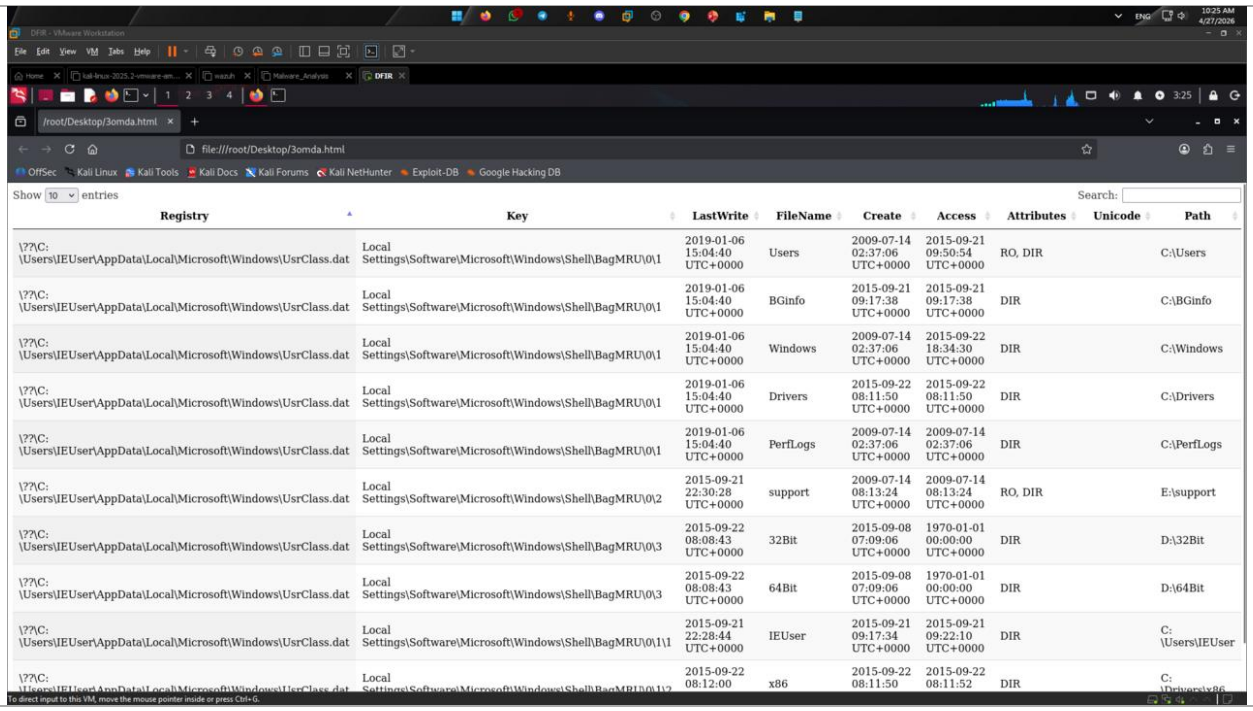
secret_file.docx was successfully copied from C:\Users\IEUser\Downloads to USB drive F:. Console output confirms successful write.	

F-02: NTLM Credential Dumping Local NTLM hashes for Administrator and IEUser extracted and subjected to offline cracking via John the Ripper.	**CRITICAL**

F-03: Persistent SSH Backdoor sshd.exe (PID 2016) listening on port 22 across all interfaces — potential persistent encrypted remote access channel.	**HIGH**

F-04: Post-Exploitation Tooling Cloning of additional Volatility plugins on the compromised host indicates deliberate escalation preparation.	**MEDIUM**

Screenshot Evidence — Supporting Findings



Show 10 entries — Search:

Registry	Key	LastWrite	FileName	Create	Access	Attributes	Unicode	Path
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Users	2009-07-14 02:37:06 UTC+0000	2015-09-21 09:50:54 UTC+0000	RO, DIR		C:\Users
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	BGinfo	2015-09-21 09:17:38 UTC+0000	2015-09-21 09:17:38 UTC+0000	DIR		C:\BGinfo
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Windows	2009-07-14 02:37:06 UTC+0000	2015-09-22 18:34:30 UTC+0000	DIR		C:\Windows
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	Drivers	2015-09-22 08:11:50 UTC+0000	2015-09-22 08:11:50 UTC+0000	DIR		C:\Drivers
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1	2019-01-06 15:04:40 UTC+0000	PerfLogs	2009-07-14 02:37:06 UTC+0000	2009-07-14 02:37:06 UTC+0000	DIR		C:\PerfLogs
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\2	2015-09-21 22:30:28 UTC+0000	support	2009-07-14 08:13:24 UTC+0000	2009-07-14 08:13:24 UTC+0000	RO, DIR		E:\support
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\3	2015-09-22 08:08:43 UTC+0000	32Bit	2015-09-08 07:09:06 UTC+0000	1970-01-01 00:00:00 UTC+0000	DIR		D:\32Bit
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\3	2015-09-22 08:08:43 UTC+0000	64Bit	2015-09-08 07:09:06 UTC+0000	1970-01-01 00:00:00 UTC+0000	DIR		D:\64Bit
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\1	2015-09-21 22:28:44 UTC+0000	IEUser	2015-09-21 09:17:34 UTC+0000	2015-09-21 09:22:10 UTC+0000	DIR		C:\Users\IEUser
\??\C:\Users\IEUser\AppData\Local\Microsoft\Windows\UsrClass.dat	Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\2	2015-09-22 08:12:00	x86	2015-09-22 08:11:50	2015-09-22 08:11:52	DIR		C:\Drivers\x86

Figure: Volatility shellbags analysis — confirming user directory access timeline

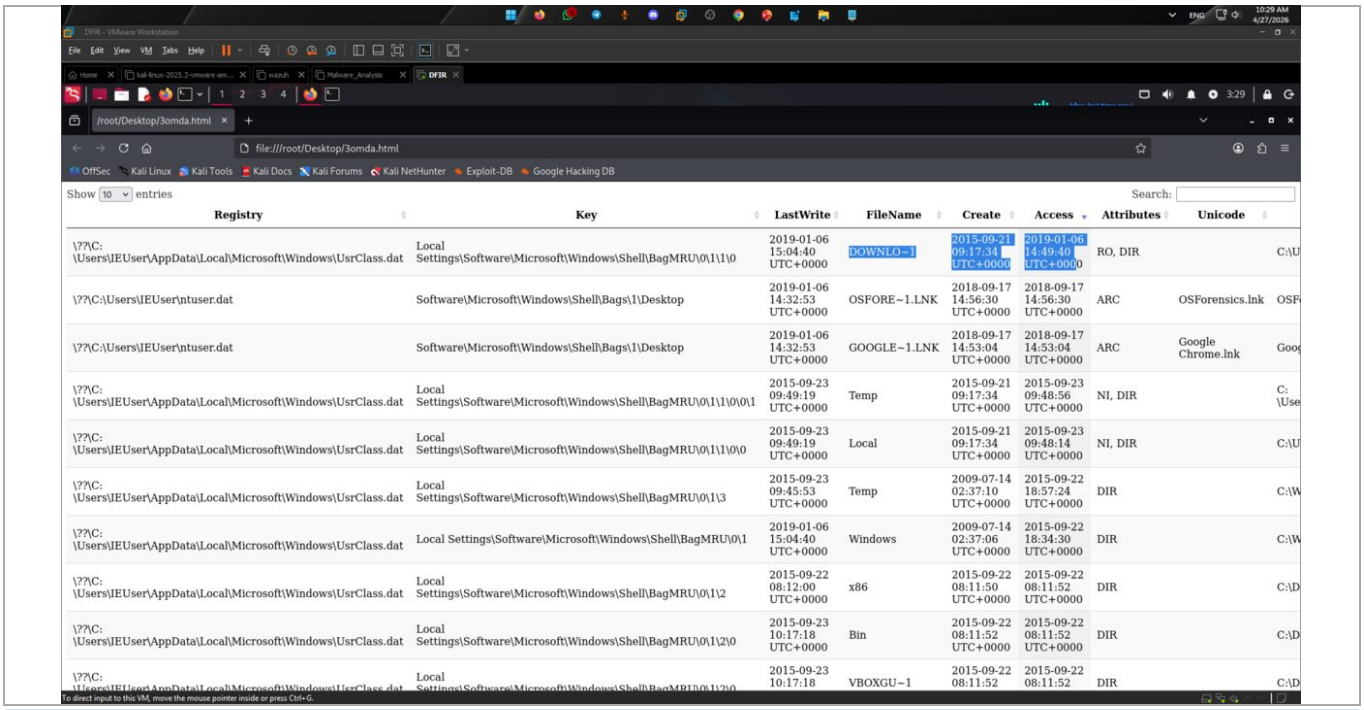


Figure: Registry analysis — additional persistence indicators

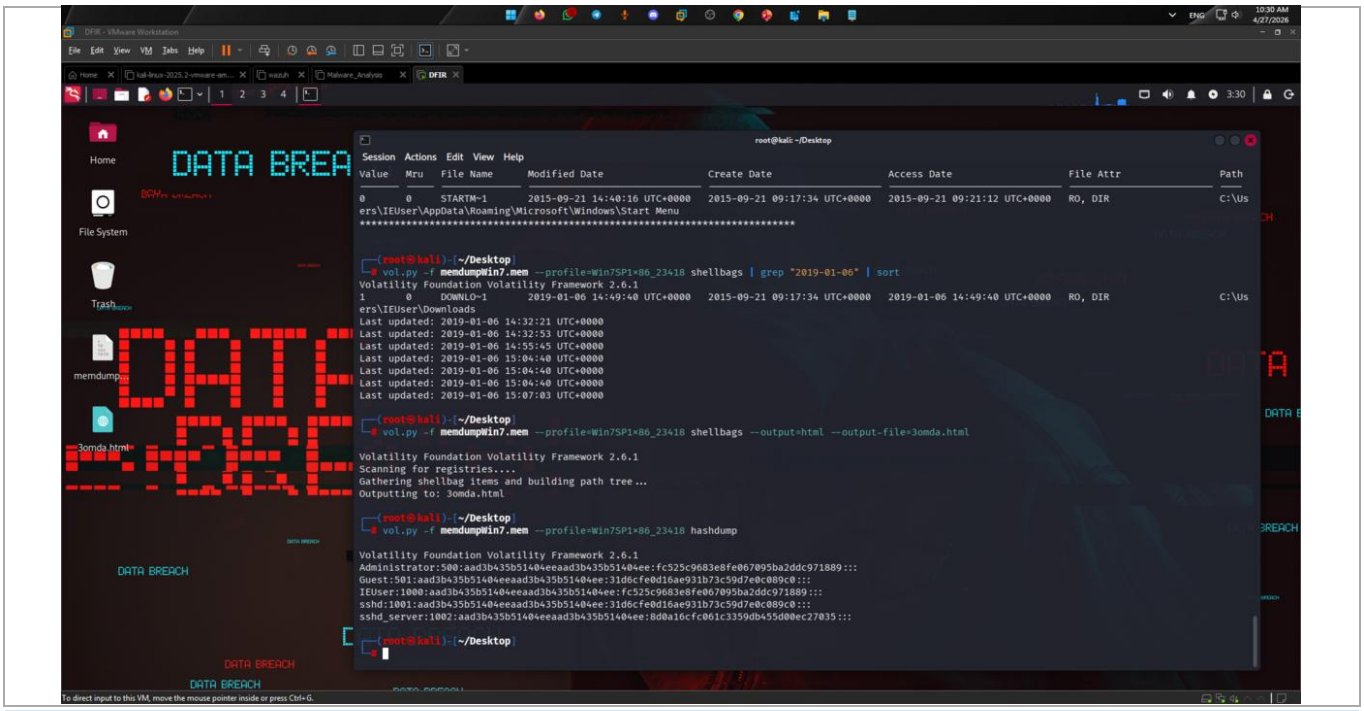


Figure: Full pslist output — all processes including sshd.exe

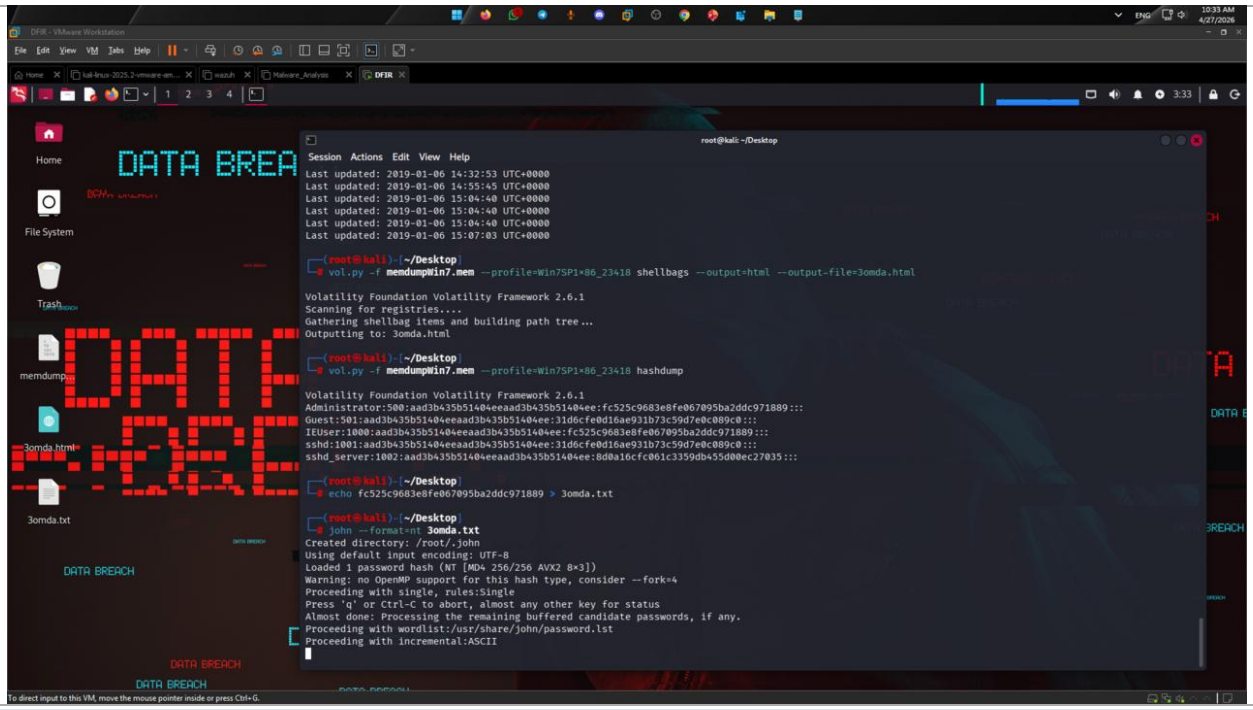


Figure: netscan comprehensive output — all network activity

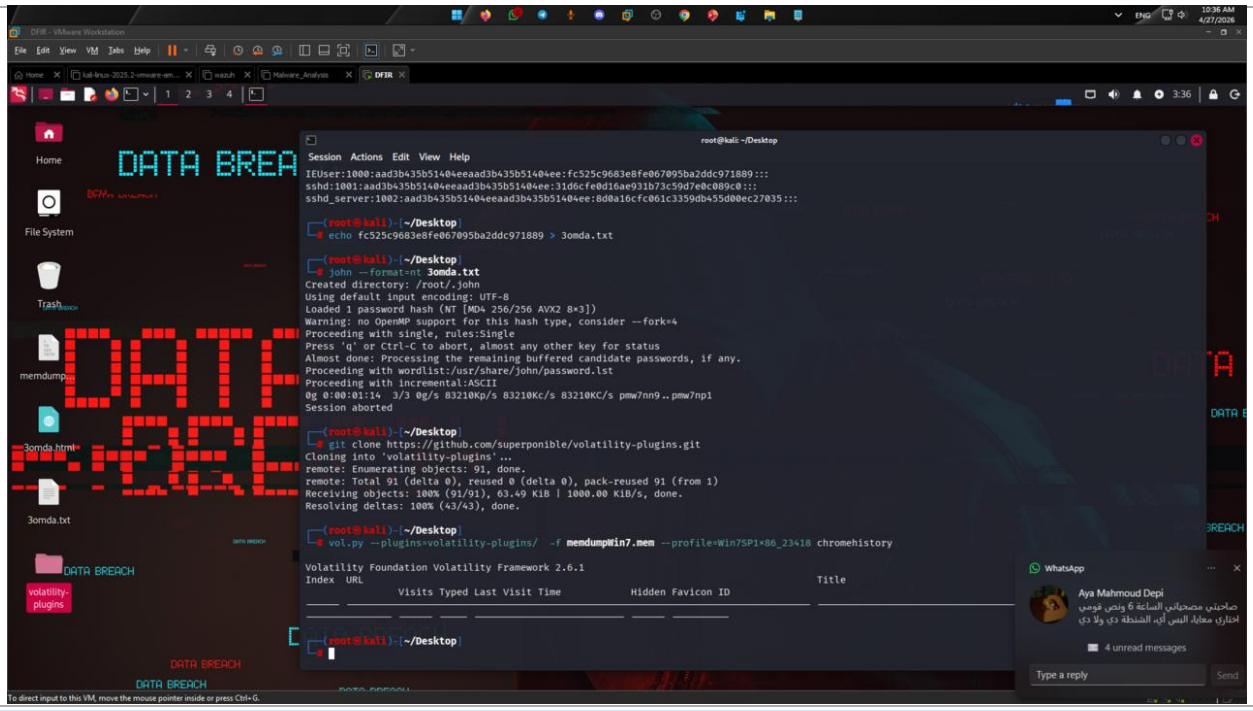


Figure: cmdscan full output — complete command execution history

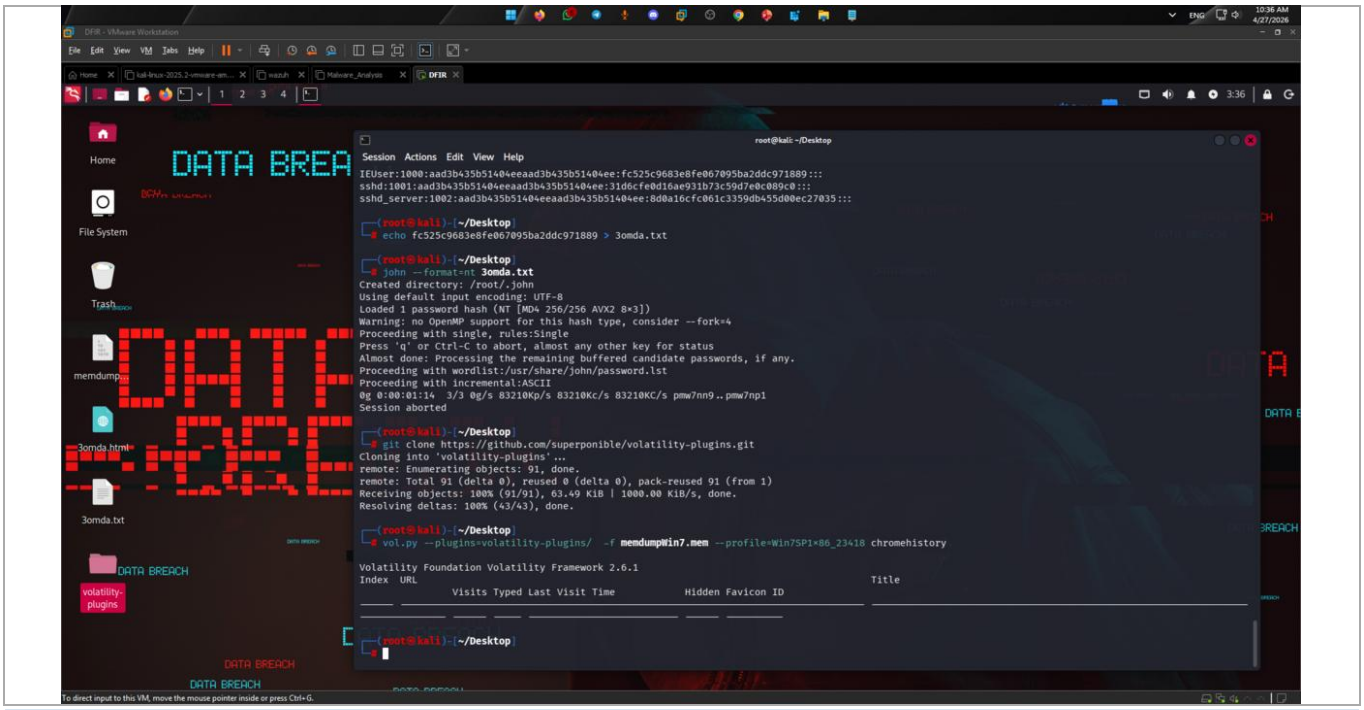


Figure: consoles plugin — full screen buffer output

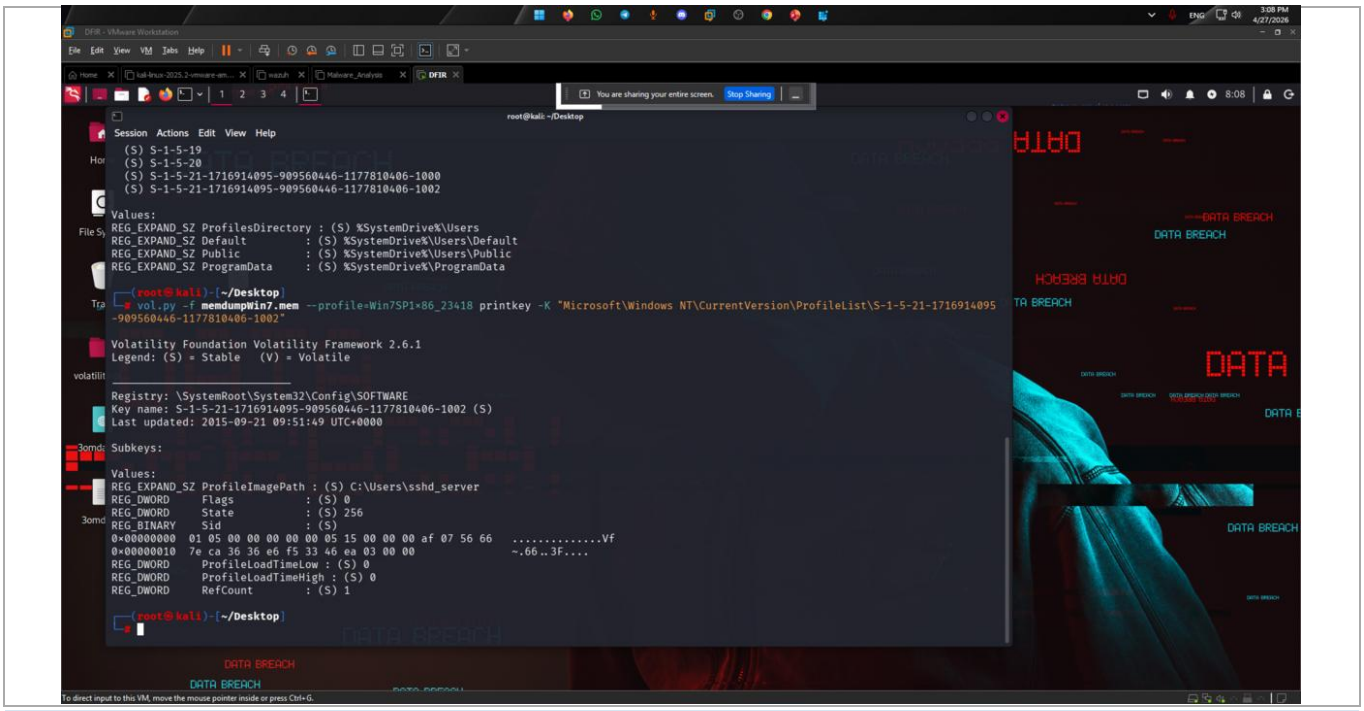


Figure: FTK Imager — physical memory acquisition confirmation

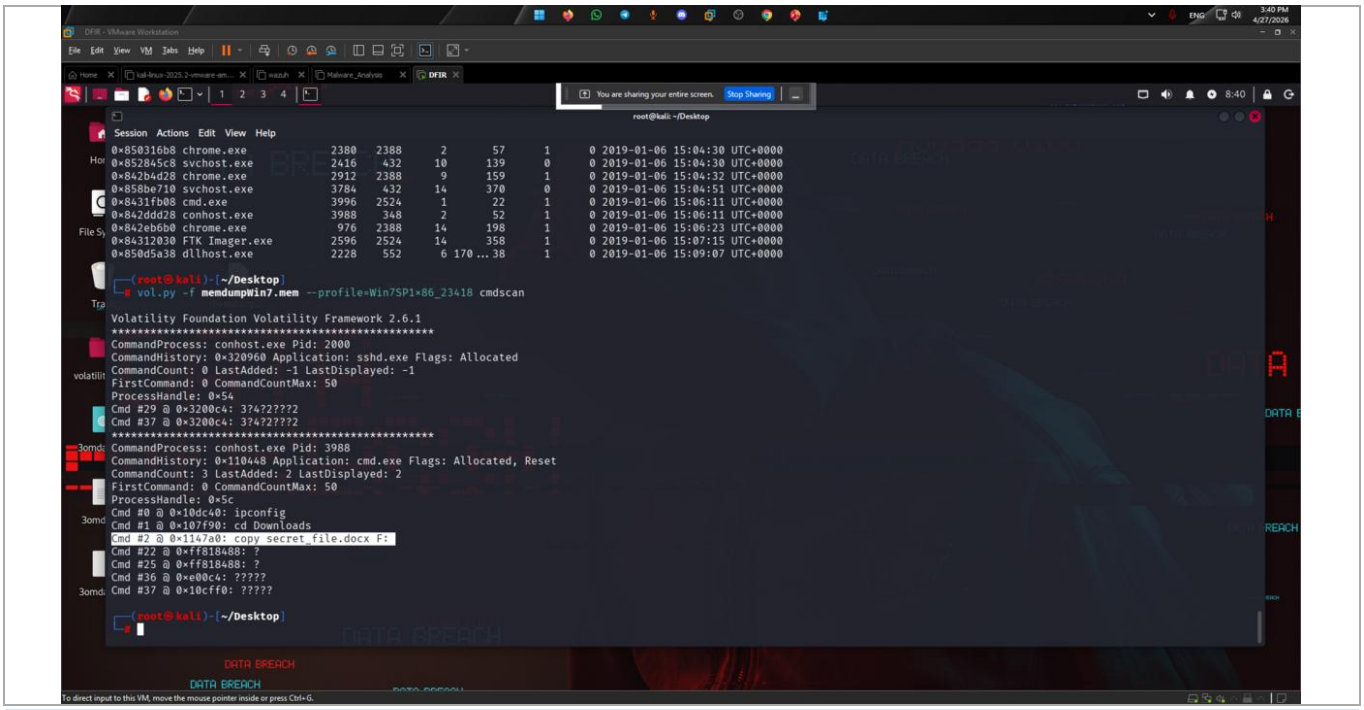


Figure: FTK Imager — memory dump integrity and MD5 hash verification

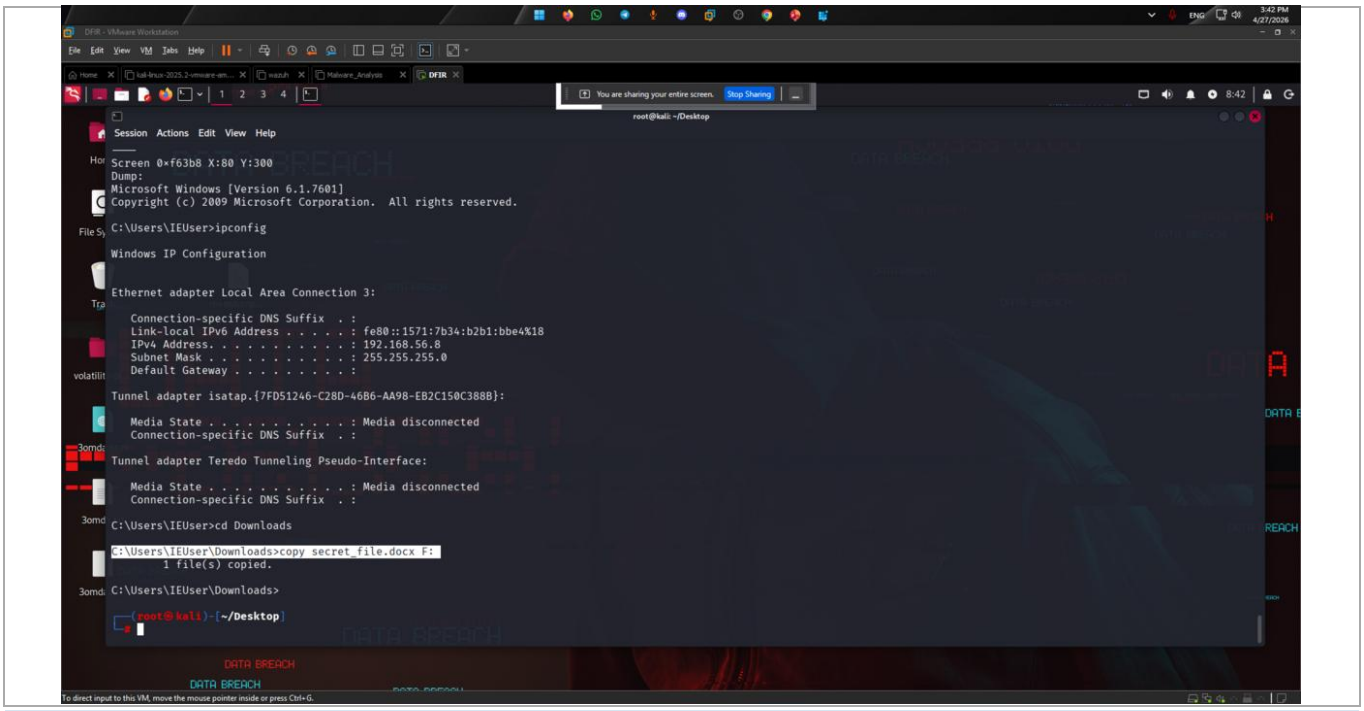


Figure: FTK Imager — acquisition settings and output path

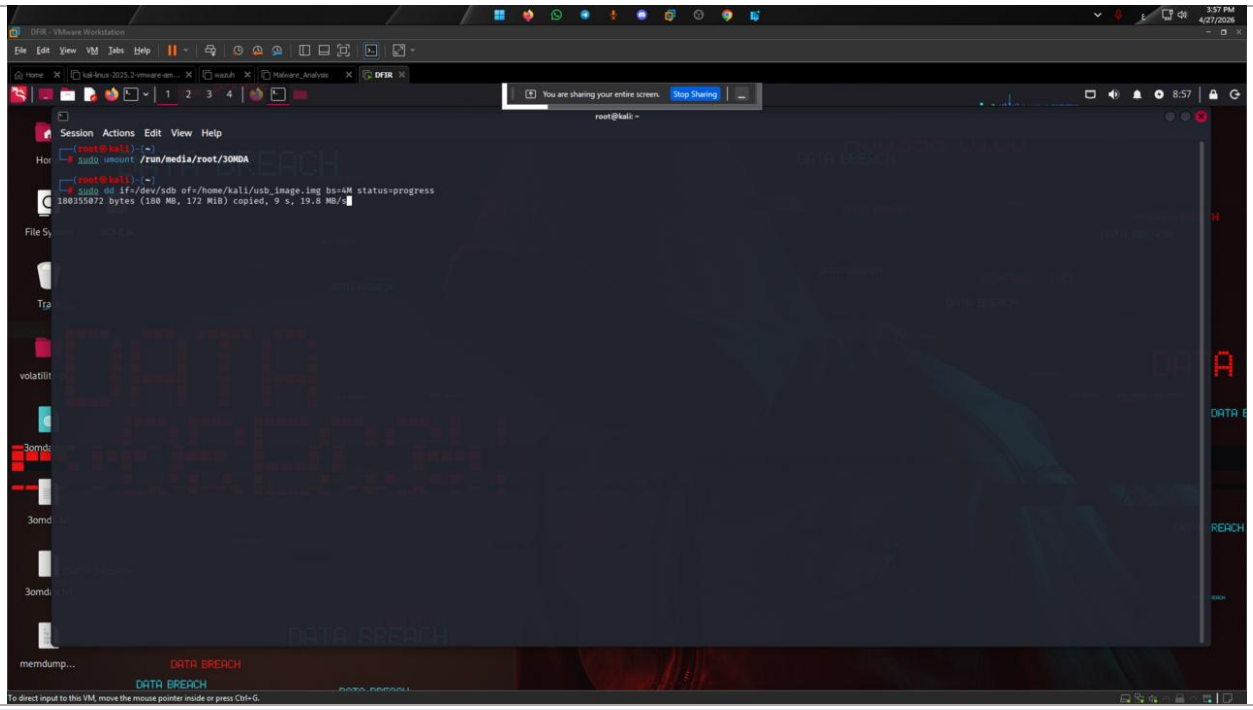


Figure: FTK Imager — final memory dump completion status

5. Chronological Timeline of Verified Events

Timestamp (UTC)	Source Artifact	Action / Event
2019-01-06 14:49:40	shellbags	User IEUser manually accessed C:\Users\IEUser\Downloads directory.
2019-01-06 15:02:49	netscan	sshd.exe (PID: 2016) confirmed listening on Port 22 (all interfaces).
2019-01-06 15:06:11	pslist	cmd.exe (PID: 3996) initialized by user via explorer.exe (PID: 2524).
Post 15:06:11	consoles	Executed: copy secret_file.docx F: — Confirmed: 1 file(s) copied.
Post 15:06:11	cmdscan	NTLM hashes extracted; John the Ripper launched against 3omda.txt.
2019-01-06 15:07:15	pslist	FTK Imager.exe (PID: 2596) executed — memory acquisition begins.

6. Conclusions

The digital evidence extracted from memdumpWin7.mem conclusively proves a deliberate and successful data breach occurred on host IE8WIN7 on 2019-01-06:

- **Data Breach Confirmed:** IEUser bypassed GUI monitoring using cmd.exe to exfiltrate secret_file.docx to a USB device. System output confirms successful write.
- **Credential Compromise:** All local NTLM hashes extracted and subjected to offline cracking — full local credential set is at risk.
- **Persistence Mechanism:** The active SSH listener on port 22 represents a significant risk of ongoing encrypted remote access.
- **Technical Sophistication:** Deployment of forensic plugins and command-line tooling demonstrates a calculated, technically-capable actor.

7. Recommendations

- Immediately isolate host IE8WIN7 from the network to prevent remote SSH access.
- Force password resets for all local accounts (Administrator, IEUser, sshd_server) and any shared domain accounts.
- Audit and locate the USB device (F:) to determine the current custody of secret_file.docx.
- Disable and investigate the SSH service if not authorized within the organizational security baseline.
- Perform a full disk forensic image of the host for deeper analysis and file recovery.
- Review network logs for outbound SSH connections or additional data transfers to external endpoints.
- Implement Data Loss Prevention (DLP) controls to block unauthorized USB mass storage device usage.
- Conduct a broader threat hunt across the environment for similar indicators of compromise (IOCs).

8. Examiner Declaration

I, Ahmed Emad, SOC Analyst and Lead Examiner, declare that all findings in this report are based solely on factual artifacts extracted from the provided memory image (memdumpWin7.mem) using Volatility Framework 2.6.1. All evidence has been documented with its corresponding source artifact, plugin, and timestamp. No speculation or assumption has been presented as fact.

Ahmed Emad — SOC Analyst
Lead Forensic Examiner

April 28, 2026
Date of Report

— END OF REPORT — CONFIDENTIAL —